Knowledge-Based Systems 164 (2019) 274–291

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

Nonstationary time series transformation methods: An experimental review

Rebecca Salles [a], Kele Belloze [a], Fabio Porto [b], Pedro H. Gonzalez [a], Eduardo Ogasawara [a,*]

[a] Federal Center for Technological Education of Rio de Janeiro (CEFET/RJ), Brazil
[b] National Laboratory for Scientific Computing (LNCC), Brazil

ARTICLE INFO

Article history:
Received 2 June 2018
Received in revised form 24 October 2018
Accepted 29 October 2018
Available online 3 November 2018

Keywords:
Nonstationarity
Time series
Transformation methods
Prediction models
Review

ABSTRACT

Data preprocessing is a crucial step for mining and learning from data, and one of its primary activities is the transformation of data. This activity is very important in the context of time series prediction since most time series models assume the property of stationarity, i.e., statistical properties do not change over time, which in practice is the exception and not the rule in most real datasets. There are several transformation methods designed to treat nonstationarity in time series. However, the choice of a transformation that is appropriate to the adopted data model and to the problem at hand is not a simple task. This paper provides a review and experimental analysis of methods for transformation of nonstationary time series. The focus of this work is to provide a background on the subject and a discussion on their advantages and limitations to the problem of time series prediction. A subset of the reviewed transformation methods is compared through an experimental evaluation using benchmark datasets from time series prediction competitions and other real macroeconomic datasets. Suitable nonstationary time series transformation methods provided improvements of more than 30% in prediction accuracy for half of the evaluated time series and improved the prediction in more than 95% for 10% of the time series. Furthermore, the adoption of a validation phase during model training enables the selection of suitable transformation methods.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Adequate data preprocessing is an important activity in any application aiming at data analytics. It generally demands a long time and dedication [1,2]. The main objective of data preprocessing is ensuring the quality of data serving as input to applied learning methods and therefore avoid obtaining inaccurate and/or incorrect results and conclusions [3]. Among the activities commonly performed during preprocessing, we can list data cleaning, feature and sample selection, outlier removal, normalization, and data transformation.

The data transformation activity becomes particularly important in the context of prediction [2,4]. This context encompasses both problems of classification (prediction of discrete data) and regression (prediction of continuous data) [2,4,5]. However, henceforth we focus on the problem of predicting numeric time series data through regression. For simplicity, we may refer to prediction and regression interchangeably in this paper.

Although a great variety of time series prediction methods exists in literature [6], many of these methods and the majority of works that handle time series assume that the available time series is stationary [7]. In a stationary time series, statistical properties, such as mean, variance and covariance, remain constant over time and in any sample of data [7,8]. However, in practice, we observe that such properties are not constant in the majority of real-world time series, especially in socioeconomics [9], where many of them are nonstationary. Thus, when observed the presence of nonstationarity in a time series, it is a usual approach to search for ways to transform them to achieve stationarity so that the known time series prediction methods can be applied.

There exist several transformation methods in literature for coping with nonstationarity in times series. However, the choice for an adequate method to a particular time series application is not a simple task. The analysis of their features and expected advantages is crucial. Some of the features that should be considered are their initial data assumptions (including different kinds of nonstationarity, linearity, and seasonality) and their intrinsic properties (mathematical transformation or computational algorithm). In this context, a thorough overview of different transformation methods for handling nonstationary time series and their respective features becomes particularly important.

Although the literature on nonstationarity in time series is gaining attention, not many authors focus on reviewing transformation methods for nonstationarity treatment [10,6]. Yang and Zurbenko

* Corresponding author.
E-mail address: eogasawara@ieee.org (E. Ogasawara).

https://doi.org/10.1016/j.knosys.2018.10.041
0950-7051/© 2018 Elsevier B.V. All rights reserved.

[10] present a brief review of nonstationary time series analysis methods (including some of their development) encompassing many problems besides prediction. Cheng et al. [6] presents a comprehensive review and a comparative analysis of models for time series of complex systems (presenting nonstationarity and nonlinearity), including multivariate and machine learning models. Nonetheless, to the best of our knowledge, there are no works that focus on the study of different ways to coerce a time series into stationarity and their effects on univariate time series prediction. This paper targets such gap by providing a review of a broader number of time series transformation methods presented in the literature to handle nonstationarity. The paper contributes by providing:

- A thorough review of nonstationary time series transformation methods for time series prediction organized in categories.
- A timeline of related works presenting the evolution of data transformation methods for nonstationary time series prediction grouped by their domain of application.
- A comparative experimental analysis of representative transformation methods for the time series prediction problem.

The goal of the comparative experimental analysis is to provide a practical point of view regarding the advantages and limitations of the reviewed transformation methods to the univariate time series prediction problem. According to the experimental evaluation conducted, suitable nonstationary time series transformation methods provided improvements of more than 30% in prediction accuracy for approximately half (130/262) of the evaluated time series. Accuracy improvements reached more than 95% for over 10% of the evaluated time series. This observed outcome suggests the need for considering these transformation methods and for comparing them during time series prediction. Additionally, the adoption of a validation phase for exploring different transformation methods generally led to selecting one of the top 5 most appropriate for a particular time series.

Besides this introduction, the remainder of this paper is organized as follows. Section 2 provides basic concepts regarding nonstationarity in time series. In Section 3, a review of the most researched transformation methods for coping with nonstationary time series for the problem of prediction is presented. Sections 4 and 5 respectively describe the reviewed transformation methods categorized by their two main classes: mapping-based and splitting-based. The following Section 6 describes other relevant techniques for modeling nonstationary times series. A timeline of publications grouped by their domain of application of the reviewed methods is presented in Section 7. Section 8 discusses the effects of different transformation methods to the problem of prediction of nonstationary time series through experimental analysis. Finally, Section 9 concludes.

2. Time series and nonstationarity

A time series is a sequence of observations of an object of interest collected over time. When observations are related to a single variable, a time series is referenced as a univariate one. Commonly, the behavior of a univariate time series is studied as a function of its own past data [11]. Generally, we may consider a univariate time series X as a stochastic process, that is, a sequence of n random variables, $\langle x_1, x_2, x_3, \ldots, x_n\rangle$, where x_1 represents the value assumed by the series at the first (oldest) time point and x_n represents the value of the series at the newest time point [4,8]. The length n of a time series X is represented as $|X|$ and a specific time series observation is referenced as, x_t, indexed in time by $t = 1, \ldots, n$.

Most methods applied for time series prediction assume that the behavior of a time series presents a level of regularity over time, which is generally approached with the study of the concept of stationarity [7,8]. The following subsections formalize the different types of stationarity.

2.1. Strict stationarity

In a strictly stationary time series, the probabilistic behavior of every possible sequence of values $\langle x_{t_1}, x_{t_2}, \ldots, x_{t_k}\rangle$ is equal to that of the time shifted sequence $\langle x_{t_{1+h}}, x_{t_{2+h}}, \ldots, x_{t_{k+h}}\rangle$. Therefore, Eq. (1) is valid for all $k = 1, 2, \ldots$, all arbitrary integer time points $t_1, t_2, \ldots, t_k$, all arbitrary numbers $c_1, c_2, \ldots, c_k$, and all possible time shifts $h = 0, \pm1, \pm2, \ldots$ [8].

$$P\{x_{t_1} \leq c_1, \ldots, x_{t_k} \leq c_k\} = P\{x_{t_{1+h}} \leq c_1, \ldots, x_{t_{k+h}} \leq c_k\} \quad (1)$$

Usually, the definition of strict stationarity is considered too strong for most applications. This is due to the fact that such definition implies that all possible distribution functions for all subsets of a time series must be in agreement with their counterparts in the shifted sequence for all values of h. A property which is scarcely observed in most time series. Moreover, when handling a single dataset, the evaluation of strict stationarity is often not straightforward [8].

2.2. Weak stationarity

A more widely adopted version of stationarity, namely weak stationarity, imposes conditions only on the first two moments of a time series. A weakly stationary time series, X, is a finite variance stochastic process such that: (i) the mean function, $E(x_t) = \mu_t = \mu$, is constant and does not depend on time t; and (ii) the autocovariance function, $\gamma(s, t)$, between x_t and the time-shifted time series value x_s depends only on the difference $|s - t|$ [7,8].

In other words, a weakly stationary time series presents constant mean and variance, and its covariance function depends only on the time difference [10]. These constraints are very important since they enable statistical inference to be drawn based on any sampled subset of a time series [11]. As in most works in literature, henceforth the term stationary refers to a weakly stationary process.

An example of a stationary time series may be visualized in Fig. 1a, where one may observe mean and variance functions which are independent of time. It represents a first order autoregressive model (AR(1)), which is defined as in Eq. (2), where α is a constant and $\omega_t \sim N(0, \sigma_\omega^2)$ [7]. In Fig. 1a, $\sigma_\omega^2 = 2$, $\theta = 0.5$, and $\alpha = 0$. Since $0 < \theta < 1$, any relevant impacts of past observations eventually become negligible and do not affect the global behavior of the time series. It follows that this model presents constant statistical properties such as $E(x_t) = \mu_t = \alpha/(1 - \theta) = 0$ and $VAR(x_t) = \sigma_\omega^2/(1 - \theta^2) \sim 2$, hence being considered stationary.

$$x_t = \alpha + \theta x_{t-1} + \omega_t, \quad 0 < \theta < 1 \quad (2)$$

2.3. Nonstationarity

If a time series X violates any of the constraints imposed by a stationary process, it is considered a nonstationary time series. Nonstationarity may manifest in many different ways. Generally, it implies that the mean and/or variance functions of a time series are non-constant and vary over time, that is, they are dependent on time t. The changes in mean and/or variance in time series are often due to deterministic trends, structural breaks, level shifts or changing variances (a condition known as heteroscedasticity). They can also be due to the presence of unit roots [11]. Fig. 1(b–e) shows representative nonstationary time series.

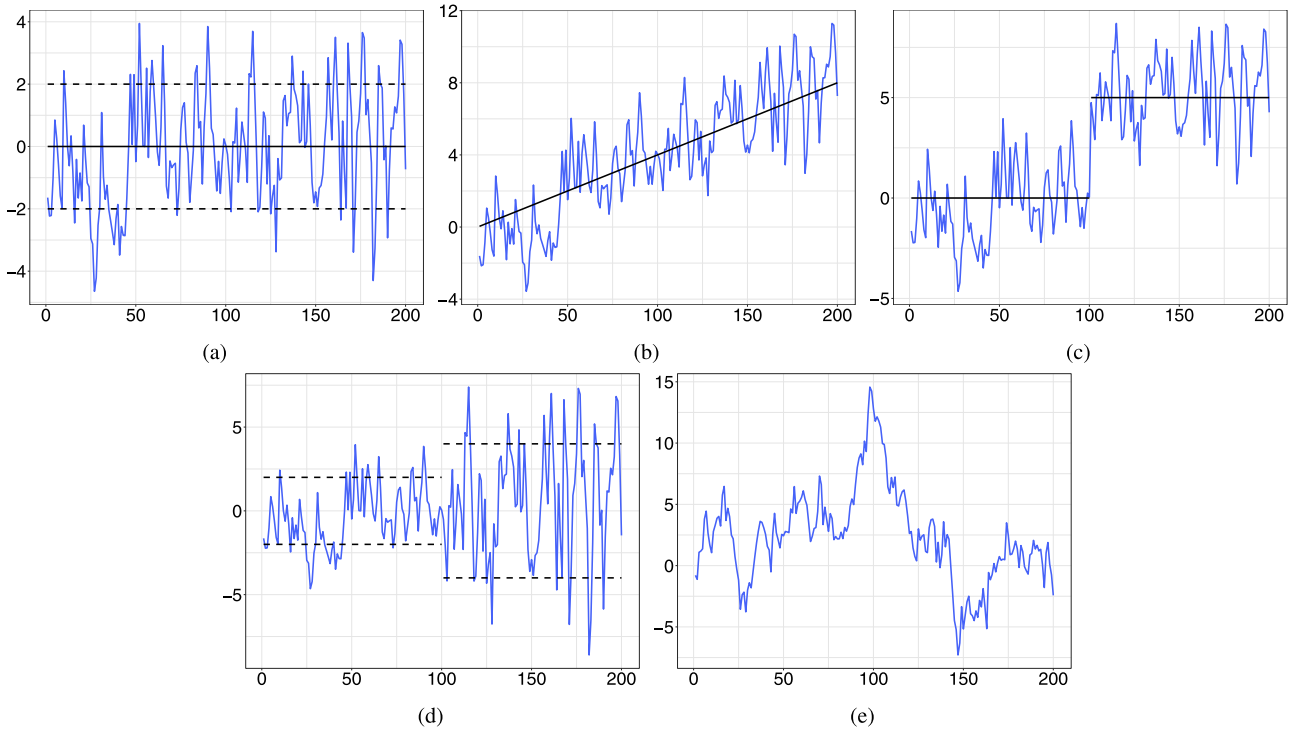


Fig. 1. Examples of time series presenting the properties of (a) stationarity, and nonstationarity in the form of (b) trend stationarity, (c) level stationarity, (d) heteroscedasticity and (e) difference stationarity. The solid and dashed black lines represent the mean and the variance functions of the time series, respectively.

A trended model might be considered the simplest form of nonstationary time series. This model represents a process that has stationary behavior around a deterministic trend. This trend shifts the mean of a time series causing it to increase or decrease over time. Commonly, the deviations of a systematic trend may be a stationary variable, known as a detrended variable, which may be analyzed instead of the original time series. In that case, usual stationary models are applicable [11,10]. A time series that presents this behavior is called trend stationary. We may write an example model of such time series by adding a deterministic linear trend to the AR(1) model presented in Eq. (2). This model is defined as in Eq. (3), where βt is the trend term and $\omega_t \sim N(0, \sigma_\omega^2)$ represents white noise. The mean function of the process $E(x_t) = \mu_t = \beta t$ varies over time, violating the stationarity constraints. The time series represented in Eq. (3) may be observed in Fig. 1b where one can see the linear increasing mean function. This model gives an example of a process which is nonstationary in mean.

$$x_t = \alpha + \theta x_{t-1} + \beta t + \omega_t, \quad 0 < \theta < 1 \tag{3}$$

Nonstationarity in a time series may also be caused by structural breaks, that happen at specific points in time, usually due to environment changes. These structural breaks may eventually result in level shifts in a time series, which cause the mean function to be different for different portions of the series. In that case, a time series can be partitioned, and one can separately analyze each data portion with different statistical properties, provided that the timing of a structural break is known [11]. Another way to handle structural breaks and level shifts in the mean function while modeling a time series is to make use of a dummy variable defined as zero before the point of a structural break and one after it. In case a time series presents local stationary properties on each different portion divided by a level shift it is known as being level stationary. An example of a level stationary time series is observed in Fig. 1c where one can see the level shifts of the mean function. The plot in Fig. 1c represents the model in Eq. (4) which is derived from Eq. (2) by the addition of a dummy variable d_t, with the level shift $\delta = 5$ and the time of the structural break $t_b = 100$. This model gives another example of nonstationarity in the mean.

$$x_t = \alpha + \theta x_{t-1} + \delta d_t + \omega_t, \quad 0 < \theta < 1, \quad d_t = \begin{cases} 0, & t \leq t_b, \\ 1, & t > t_b. \end{cases} \tag{4}$$

Another cause of nonstationarity which results from structural breaks is the change in variance over time, a condition which is commonly known as heteroscedasticity [11,8]. Heteroscedasticity arises from environment changes that make the volatility of time series observations increase/decrease over time. Time series which present this condition are called heteroscedastic. Analogous to level shifts, the different variance properties in different portions of a time series can be addressed by partitioning the series or by modeling the changes in variance with a dependency on the structural breakpoints. An example of heteroscedastic time series is depicted in Fig. 1d where different variance properties on the first and last portions of the series are easily observable. The series presented in Fig. 1d represent the same model defined in Eq. (2) and the same series of Fig. 1a, but in this case, ω_t is set as $\omega_t \sim N(0, \sigma_\omega^2 = 2)$ for $t = 1, \ldots, 100$ and $\omega_t \sim N(0, \sigma_\omega^2 = 4)$ for $t = 101, \ldots, 200$. This model gives an example of a time series which presents nonstationarity in the variance.

An important type of nonstationarity, which in many cases is observed in real-world series, is caused by the presence of a unit root in the characteristic polynomial of a time series model. Without a unit root, time series observations tend to fluctuate around deterministic components such as a mean or a trend. Conversely, when a unit root is present, observations do not revert to a historical level and may wander in any direction. The presence of a unit root implies that the time series suffer from the influence of long-run components or stochastic trends [11]. In that case, the removal of a stochastic trend, usually done by the application of a process called differencing, is often helpful to coerce such time series to stationarity. For that reason nonstationary time series that present unit roots are also known as difference stationary [12]. A difference

stationary time series is presented in Fig. 1e that represents the so-called random walk model, which can be formulated as in Eq. (5). This model assumes that the value of a time series x_t at a time t can be explained by the value of the series at the time $t-1$ plus a random movement represented by ω_t [8].

$$x_t = \alpha + x_{t-1} + \omega_t = \alpha + \sum_{i=1}^{t} \omega_i \tag{5}$$

The random walk model in Eq. (5) is also derived from the AR(1) model in Eq. (2) by defining $\theta = 1$. The definition of $\theta = 1$ implies that any impacts caused by past observations result in a permanent effect on the global behavior of a time series. In this case, the mean function of the process is not fixed, and the variance function tends to increase over time [11] as can be observed by Eq. (6). This model gives an example of a process which presents a unit root and is nonstationary both in mean and in variance [10].

$$VAR(x_t) = VAR(\alpha) + \sum_{i=1}^{t} VAR(\omega_i) = t\sigma_\omega^2 \rightarrow \infty \; as \; t \rightarrow \infty \tag{6}$$

It is also important to remark that frequently the dependence of a time series on past data may occur by multiples of some underlying seasonal lag S. In that case, a time series presents periodic components, and therefore, its statistical properties such as mean and variance may periodically change, creating a dependence on time t. This makes seasonality another particular form of nonstationarity, which is often found in time series [10].

Generally, any form of nonstationarity, if not properly addressed, can have a relevant impact on time series prediction applications. Overlooking nonstationarity properties in a time series may lead to misleading statistical inferences and bad or unexpected prediction results. The next section presents a categorization for some of the most researched transformation methods prepared for application to nonstationary time series.

3. Nonstationary time series transformation methods

Nonstationarity poses challenges to time series prediction, especially since this property is pervasive in many real-world scenarios. As a consequence, several methods for statistical analysis of nonstationary time series have been developed [13]. In this section, we discuss some of the most researched time series transformation methods for handling nonstationarity. We focus on the methods generally applied to the problem of time series prediction. For providing a better overview of the methods and for helping the discussion of their particular features we introduce a general categorization presented in Fig. 2.

3.1. Mapping and splitting transformations

The diagram presented categorizes 18 of the most researched transformation methods for handling nonstationarity in time series by their two main classes: (i) mapping-based and (ii) splitting-based.

The mapping-based transformations correspond to methods that map the data of a time series to another representation. They return a new time series generated by passing each of its observations through a mathematical process. The transformed time series usually presents interesting properties, including stationarity, that are useful for the application of time series prediction methods that take them as input. The same parameters of the undergone transformation process may afterward be used for reversing it and obtaining time series data consistent with the original time series representation.

The splitting-based transformations refer to methods that are also able to produce a new representation of time series data. These methods apply different techniques for splitting a time series into a number of component series. Each component series generally serve as simpler or even stationary inputs for time series prediction methods that can analyze and predict them separately [14]. The undergone decomposition process may afterward be reversed, generally by an additive or multiplicative combination of the component series, to obtain time series data consistent with the original representation. Splitting-based transformation methods are drawing increasingly more attention during the last few decades, especially when associated with computational intelligence techniques [15]. However, it might still be considered under-researched [13,10].

3.2. Parametric and nonparametric methods

Within these two main classes, we can also categorize the methods into parametric and nonparametric according to their initial assumptions regarding the underlying behavior of a time series. Parametric methods make relatively more rigorous and inflexible assumptions about the parameters of the data distribution of a time series, whereas nonparametric methods are data-driven and generally try to infer the number and nature of the parameters from the input time series [2].

From another view, we can define parametric methods as having a fixed number of data parameters, while nonparametric methods allow for the number of parameters to grow with the amount of data in the training set. Although we can say that parametric methods make stronger assumptions about the nature of the data distributions of a time series, they are generally faster, straightforward to use, and easier to interpret. Moreover, although nonparametric methods present more flexibility, they can often demand an infeasible computational cost for large time series datasets and are prone to overfit the data [16].

Some parametric methods practice trend estimation. The estimated trend of a time series may be deterministic or stochastic. A deterministic trend is estimated as a deterministic function over time, which can be modeled by a standard regression model or a model of sine and cosine curves when a time series presents hidden cycles at different frequencies. Conversely, a stochastic trend is modeled as a nondeterministic function over time, such as an autoregressive moving average (ARMA) model, which usually serves as a stochastic component [10].

3.3. General settings and time series properties

The categorization presented in this paper also classifies the transformation methods for handling nonstationary time series according to a set of features that may often be of interest to researchers, such as their class of implementation, the kinds of nonstationarity and the time series properties considered by the method. The implementation classes of the methods, being either a mathematical transform or a computational algorithm, are represented by the different geometric icons associated with each method. Some of the methods involve a time series transform that is performed by computational algorithms. In that case, their icons are formed by two of the respective geometric symbols.

The kinds of nonstationarity discussed in Section 2.3, namely nonstationarity in mean and in variance, which the transformation methods are prepared to handle, are also represented in the categorization diagram by the background color of the icon of each method. Furthermore, some of the reviewed methods are also prepared to handle often crucial time series properties such as nonlinearity and seasonality. These methods and the properties they consider are represented in Fig. 2 by the patterns (horizontal/vertical lines) which fill their respective icons.

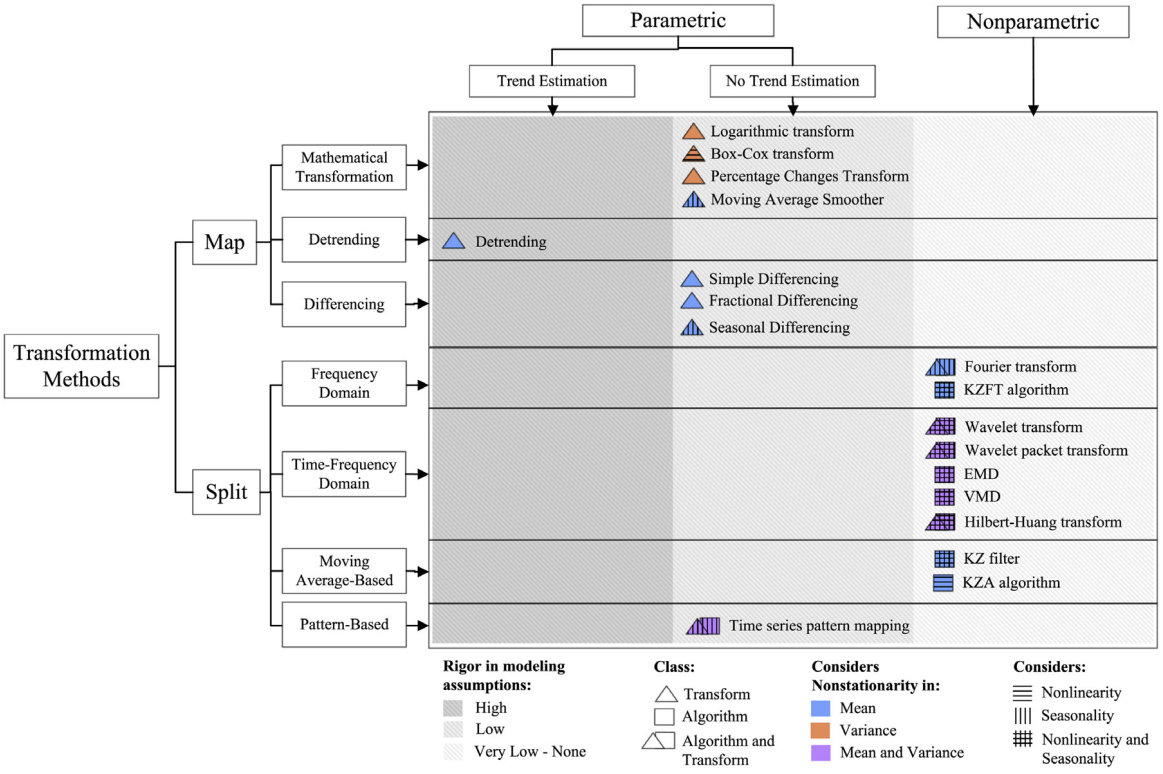


Fig. 2. Categorization diagram for some of the most researched transformation methods for coping with nonstationarity in univariate time series prediction.

4. Mapping-based transformation methods

Mapping-based transformation methods derive new mapped representations of a time series usually by undergoing one of three processes: (i) basic mathematical transformation, based on the use of standard mathematical operations and manipulation; (ii) detrending, based on the removal of a deterministic trend; or (iii) differencing, also based on the removal of a trend, but without the need for model estimation. The most commonly used mapping-based transformation methods are described in the next subsections.

4.1. Basic mathematical transformations

Some of the most simple and the most applied nonstationary time series mapping-based transformations are the logarithmic transform (LT), Box–Cox transform (BCT) [8,17], percentage change transform (PCT), and moving average smoother (MAS) [18,3]. The methods LT, BCT, and PCT are used for minimizing or even normalizing the variability over a time series. LT tends to suppress fluctuations that occur over portions of a time series which present higher values [8], and its simple formula can be seen in Eq. (7), where $\hat{x}_t$ is the transformed version of the original time series value x_t. Often in macroeconomics, a time series is natural log-transformed (setting $b = e$) to minimize effects of nonstationarity and heteroscedasticity (non-constant variability [7]), and also to induce symmetry and normality. The estimated coefficients of a logged time series can also be interpreted as elasticities. These advantages make of natural logs one of the most applied LT [18].

$$\hat{x}_t = \log_b x_t \tag{7}$$

The BCT, as in Eq. (8), presents a more objective method for determining a suitable power transformation of a time series [19]. However, it demands the calculation of the numeric argument λ that is related to the data distribution function, which is not always known. Additionally, it can only be applied to positive valued data [17].

$$\hat{x}_t = \begin{cases} \left(x_t{}^{\lambda} - 1\right)/\lambda, & \lambda \neq 0, \\ \log x_t, & \lambda = 0. \end{cases} \tag{8}$$

The PCT assumes that there is a considerable stability in the relative percentage of change between two following observations of a time series [17]. In that case, Eq. (9) is true if the percentage change is restricted to an interval $[-100\rho, 100\rho]$, with $0 \leq |\rho| < 1$ being a small acceptable threshold.

$$\hat{x}_t \approx \log\left(\frac{x_t}{x_{t-1}}\right) \tag{9}$$

Finally, MAS has been widely used especially in finance and econometrics. It is useful for highlighting seasonality and long-term trends in a time series [8,3]. MAS can detect the evolving behavior of a time series by minimizing random noise [3], and can also be used for seasonal adjustment of a time series [18]. One of the simplest forms of the MAS is represented by Eq. (10), where k is the order of the moving average.

$$\hat{x}_t = \frac{1}{k}\sum_{i=t}^{t+k-1} x_i, \quad 1 \leq t \leq n-k+1 \tag{10}$$

4.2. Detrending versus differencing

It is also important to mention the transformations of detrending (DT) (or trend removal) and simple differencing (DIF), which are widely used in combination with time series modeling techniques. The DT transformation involves the determination and removal of an inherent trend observed in a time series behavior [8]. Generally, the observed trend is estimated and defined by a deterministic functional form, which may be represented as a fixed

component in a time series model [20]. A general DT transformation can be seen in Eq. (11), where η_t is the determined trend component and, in this case, $\hat{x}_t$ represent the variability series, that is, the residue of the time series data after trend removal. However, this method usually assumes that the deterministic trend is always appropriate over time, which is frequently not the case in many applications, especially involving nonstationary processes [17]. To overcome this drawback, recently there have been efforts towards data-driven, adaptive methods for determining trends in nonlinear and nonstationary time series [20].

$$\hat{x}_t = x_t - \eta_t \tag{11}$$

The DIF method brings some advantages compared to DT since it does not demand a parameter estimation process and is capable of generating a stationary time series when it presents stationary behavior around a deterministic or stochastic trend [8]. A first DIF transformation, simply written as in Eq. (12), can eliminate a linear trend, a second DIF eliminates a quadratic trend, and so on. For DIFs of higher-order, the backshift operator (B) is used, as seen in Eq. (13), where ∇^d is the dth differencing and the operator $(1-B)^d$ is adapted for higher orders of d [8].

$$\hat{x}_t = \nabla x_t = x_t - x_{t-1} \tag{12}$$

$$\nabla^d = (1-B)^d,\ B^k x_t = x_{t-k} \tag{13}$$

We also note the variations of DIF, the fractional differencing (FDIF) and the seasonal differencing (SDIF). The FDIF method is a similar process to DIF. It allows for the possibility of d to assume the powers $-0.5 < d < 0.5$ [8] and is commonly used for long memory time series analysis [21–23]. In the SDIF method, d assumes the value corresponding to the seasonal period s, and it can be denoted as $\nabla^s x_t = x_t - x_{t-s}$ [17]. However, a disadvantage of DIF is that the method does not produce an estimation of the inherent stationary process of a time series. In case this estimation is essential, DT may be more suited.

The DIF method can be used together with the linear ARMA models producing one of the most important time series linear prediction models, the autoregressive integrated moving average (ARIMA) models [12], which are able to model stochastic trends [17]. An ARIMA(p, d, q) model is composed of an autoregressive (AR) and a moving average (MA) modeling processes (represented by p and q, respectively) with the application of a preliminary DIF process (I) (represented by the order d) so as to handle nonstationarity in time series [7]. ARIMA models assume that an observation of a time series, x_t, can be described as a function of its p past values and its q past white noise values [8]. The latter are represented as ω_t, a Gaussian white noise series with mean zero and variance σ^2_ω. Let $\theta(B)$ and $\phi(B)$ be, respectively, the AR and the MA operators. The ARIMA model is denoted in Eq. (14) [12,8]. Variations of the ARIMA models may arise by introducing AR and MA polynomials that identify a dependence on some underlying seasonal lag, generating the seasonal ARIMA (SARIMA) models [8]. It is also possible to allow d to take non-integer values, thus performing a preliminary FDIF process (FI), generating the autoregressive fractionally integrated moving average (ARFIMA) models [24].

$$\theta(B)(1-B)^d x_t = \phi(B)\omega_t. \tag{14}$$

5. Splitting-based transformation methods

Most commonly, splitting-based transformations are nonparametric and have its origin in signal processing techniques such as spectral (frequency domain) analysis [25,13]. They decompose a time series into components (signals) having different scales (frequencies), which may be caused by different physical phenomena, to capture the intrinsic dynamics of a time series [26,10,27–29]. For example, a time series can be decomposed into short-term (high-frequency), seasonal, and long-term (low frequency) components. An advantage provided by this decomposition process is that the explanation of only a few kinds of signal components is generally simpler and more physically meaningful than a collection of estimated model parameters [8]. Moreover, the derived components are more easily modeled, which can simplify the time series prediction problem [14]. Usually, decomposition may be achieved in a frequency only or a time–frequency domain. The latter enables the preservation of information regarding localized changes. Special cases of time series decomposition are based on moving average iterations or pattern mapping (i.e., deriving patterns of a time series behavior to simplify its prediction and filter trends and long-term variations [14]). Some of the most frequently used splitting-based transformation methods are described in the following subsections.

5.1. Frequency domain decomposition

Time series analysis in the frequency domain are often based on Fourier transforms (FT) [30]. The FT creates a frequency-based representation (a spectrum) of a time series in terms of Fourier basis functions. The FT of a time series X, in this case represented as a function of time $x(t)$, can be formulated as in Eq. (15), where $F(\xi)$ represents the Fourier spectrum, ξ is a frequency component and j is the imaginary number $j = \sqrt{-1}$ [25].

$$F(\xi) = \int_{-\infty}^{+\infty} x(t) e^{-j\xi t} dt,\quad e^{-j\xi t} = \cos \xi t - j \sin \xi t \tag{15}$$

As well as for the Fourier series, the drawback of FT is that it is a global representation of a time series, and therefore information associated with time is lost [25,30]. This fact makes the FT mainly suitable when a time series does not present a relevant change in behavior over time, that is when it is stationary. Furthermore, FT does not perform well when handling large data, being limited to an interval between 0 and 2π observations [30]. In order to minimize these limitations, a windowed variation of FT, namely the windowed Fourier transform (WFT), was devised. The WFT is suitable to analyze nonstationary time series with slowly changing behavior. Also, derivations of WFT are possible such as the Kolmogorov–Zurbenko Fourier transform (KZFT), which applies the WFT with a unique short-time window, being capable of providing the best possible resolution in the frequency domain [10]. Nonetheless, WFT and its derivations may still generate time ambiguity, due to the infinite support of the Fourier wave bases [25]. Recently, a method capable of efficiently analyzing localized changes in time series has been widely applied, which is based on finite wavelet bases [25].

5.2. Time–frequency domain decomposition

Wavelets are finite basis functions localized in both time and frequency. The wavelet transform (WT) decomposes a time series (signal) by correlating it with a family of wavelets, providing an extremely flexible time–frequency representation [25,29]. The WT decomposes a time series X, which is again regarded as a function of time $x(t)$, into the wavelet series $\hat{x}(t)$ observed in Eq. (16) [30]. Here the components $\zeta(t)$ and its coefficient b represent the scale part of the wavelet series (being responsible for modeling trends and seasonality), whereas the components $\psi_{(t)}$ and its coefficient c represent the detail part of the wavelet series (corresponding to

noise or random deviations), at scale (decomposition level) l and position k. Also, L is the defined maximum decomposition level.

$$\hat{x}(t) = \sum_{k=1}^{n} b_{l,k}\zeta_{l,k}(t) + \sum_{l=1}^{L}\sum_{k=1}^{n} c_{l,k}\psi_{l,k}(t) \tag{16}$$

Since finite wavelets are usually irregular and asymmetric [30], the WT is particularly suited for analysis of nonlinear, noisy and nonstationary time series, with rapidly changing behavior [25,31,32,29]. If the wavelet transform is applied to a nonstationary time series, the resulting decomposed series usually present a better behavior than the original series, being more easily and accurately predicted, even by simple linear models like ARIMA. Predictions of the original time series with increased accuracy can then be obtained by applying the inverse WT to the predictions of the decomposed series [33,30]. WT is also a tool for multiresolution analysis (MRA), that is, modeling frequency behavior of a time series at multiple time resolutions, which is an improvement over WFT [25]. The MRA often makes of WT a more appealing option than standard econometric models [29]. The WT, in MRA, can generally be implemented as a continuous wavelet transform (CWT) or a discrete wavelet transform (DWT) [33,31]. Prediction tasks, however, require a variation of DWT, called the maximum overlap discrete wavelet transform (MODWT) [13]. We also note the derived method of discrete wavelet packet transform (WPT), which reapplies the WT to a resulting decomposed series that does not present a required level of stationarity or decorrelation [34].

Besides the WT, there are other MRA techniques that are very recent and prominent, namely the empirical mode decomposition (EMD) and the derived variational mode decomposition (VMD) [29]. Similarly to the WT, the EMD is an algorithm for nonlinear and nonstationary time series decomposition, generating a time–frequency representation of the series [29,35]. EMD is also capable of decomposing a time series into more stable components (intrinsic mode functions (IMFs)), which can be more easily modeled, improving prediction accuracy [36]. However, the advantage of EMD is that it does not depend on predetermined (wavelet) functions, since its basis functions are derived directly from the time series, therefore being adaptive and completely data-driven [29,35]. Conversely, EMD is recursive and does not perform well when it comes to separating time series components with similar frequencies. In order to overcome this limitation, the VMD algorithm was devised, which transforms the time series decomposition into a non-recursive and variational model [29,36]. The VMD is superior to EMD in the sense that it decomposes a time series more precisely and is more robust to noise [36].

Another recent noteworthy approach is presented by the Hilbert–Huang transform (HHT), which combines the use of EMD with the application of the Hilbert transform for each of the resulting IMFs [37]. The Hilbert transform is used for computing instantaneous frequencies enabling a frequency–time–energy (or amplitude) representation of a time series. The HHT is capable of accurately representing local time and frequency characteristics of nonlinear and nonstationary time series [37].

5.3. Moving average-based decomposition

Finally, we mention another example of a method for time series frequency decomposition, denominated Kolmogorov–Zurbenko filter (KZF) [10]. The Kolmogorov–Zurbenko filter (KZF) is an algorithm based on iterations of a moving average filter with noise suppression. KZF results in long-term and short-term components, where the latter is generally fit well by an ARMA model. It is also resilient to non-equally collected and/or missing time series observations [10].

It is also possible for a time series to suffer from breaks in behavior over time, a characteristic that contributes to the presence of nonstationarity. Such breaks may be analyzed by a process of intervention analysis, which is a special application of time series decomposition. This process is performed by the nonparametric Kolmogorov–Zurbenko adaptive algorithm (KZA) by zooming in the KZF results in the areas near a break. The KZA is capable of detecting the time and size of the breaks in a time series even when it presents a high level of noise. It also supports break detection when the long-term component of a time series presents a nonparametric nature, which would cause traditional statistical methods to fail [10].

5.4. Pattern-based decomposition

A special case for time series decomposition is presented by the Time series pattern mapping (PM). PM is a preprocessing technique applied to simplify the prediction of time series presenting multiple seasonalities [14]. PM decomposes a time series into patterns of its behavior, which can eliminate the properties of nonstationarity in mean and in variance. It can also remove trend and seasonal cycles of long-term periods. The decomposed patterns of a time series may be used as input to computational intelligence techniques [14].

6. Other relevant techniques

When researching methods for nonstationary time series transformation, one may also come across other relevant and useful techniques for addressing nonstationarity in time series. These techniques rely on the use of nonlinear and nonstationary modeling of time series. In particular, there is a category of models which are designed for modeling a time series by describing it as statistical terms. Some of these terms directly model nonstationary behavior, such as an inherent trend or seasonality. These models may also be combined with the application of transformations such as detrending and/or differencing to achieve stationarity. Examples of models falling into this category are set by the structural models (SM), linear mixed model (LMM), Holt–Winter's exponential smoothing (HW), Taylor's double seasonal Holt–Winter's model (THW), exponential smoothing state space model (ETS), theta forecasting (TF), and the generalized autoregressive conditional heteroscedasticity model (GARCH).

Firstly, the SM is a representative model in this category. SM consist of terms, such as trend and seasonal. These terms may provide straightforward interpretation of a time series behavior. A general SM is presented in Eq. (17), where η_t is the trend term, χ_t is the cycle term, and ω_t is the error term. Trend terms may be stochastic and can be estimated using an ARIMA process.

$$x_t = \eta_t + \chi_t + \omega_t \tag{17}$$

SM models may also be considered special cases of state space models and can be represented as so [38]. Their parameters can, therefore, be estimated by a Kalman filter [38,10]. A basic example of a state space model is given in Eq. (18), which consists of two parts: a so-called state equation (Eq. (18a)) and an observation equation (Eq. (18b)). The latter is added since $\dot{x}_t$ (in this case considered a state vector) is assumed not directly observable. Instead one can only observe x_t, which is a linear transformation of $\dot{x}_t$ with the addition of noise. Here A_t is a measurement matrix, ϵ is a coefficient matrix, and ω_t and υ_t represent noise [8].

$$\dot{x}_t = \epsilon\dot{x}_{t-1} + \omega_t, \tag{18a}$$

$$x_t = A_t\dot{x}_t + \upsilon_t \tag{18b}$$

It is possible for time series data to present dependencies and serially correlated errors due to clustering, that is when observations can be divided into related subgroups (clusters). The LMM

models belong to the category of mixed-effects models and are designed to address such dependencies among observations, which would normally make standard linear models inappropriate [39]. It considers both fixed (deterministic) and random (stochastic) effects of time in a time series. In this case, fixed effects are often modeled using a deterministic trend model, and random effects are modeled by stochastic AR or ARMA structures [10].

There are several forecasting frameworks based on the classical model of exponential smoothing. A general exponential smoothing model is expressed as in Eq. (19), where each time series value x_t, here obtained by the function of time $x(t)$, is based on the fitting function vector for the time series $f(t)$, the coefficient vector $\kappa(t)$ (T is the transpose operator) and a white noise $\omega(t)$. The function $f(t)$ depends on its past values and the transition matrix M [40].

$$x(t) = \kappa(t)^T f(t) + \omega(t), \quad f(t) = Mf(t-1) \tag{19}$$

Methods based on the exponential smoothing are applied in numerous applications due to their robustness and accuracy [24]. Among them, we note the HW method, which can be used for nonlinear modeling of heteroscedastic time series [14]. It estimates level, slope and seasonal terms at each time point [35,14]. Despite taking a long processing time to determine a few parameters, it is able to represent trend, seasonality, and randomness in an effective way [41]. The THW adapts the HW to accommodate a second seasonality [24]. The ETS model is based on an extended range of exponential smoothing, and it refers to its three basic terms: error, trend, and seasonality [24]. The TF applies differencing and exponential smoothing altogether [24].

Another model commonly applied in finance for estimating heteroscedastic time series is the GARCH [13]. A time series may be explained as a GARCH(p, q) model by Eq. (20a), where μ_t is a mean function. Particularly, the noise series ω_t is i.i.d. N(0, 1), so that the conditional distribution of the residual time series $\tilde{X}$, where $\tilde{x}_t = x_t - \mu_t$, is N(0, σ_t^2) [42,43]. The conditional variance σ_t^2 is defined by Eq. (20b) with α and β as coefficients [17]. Although GARCH models present a useful tool for treating nonconstant variability in time series, they are known for not being able to capture long memory properties and highly irregular behavior [44].

$$x_t = \mu_t + \sigma_t \omega_t \tag{20a}$$

$$\sigma_t^2 = \alpha_0 + \sum_{j=1}^{p} \alpha_j \sigma_{t-j}^2 + \sum_{j=1}^{q} \beta_j \tilde{x}_{t-j}^2 \tag{20b}$$

Finally, a noteworthy approach for addressing nonstationarity is through time-varying modeling. This approach is adopted by the Fourier nonlinear ARMA models (FNLARMA). The FNLARMA basically estimates an ARMA model with deterministic time-dependent coefficients approximated by a Fourier series [13].

7. Research scenario

The available literature on transformation methods for nonstationary time series was analyzed through a systematic mapping study using a query string involving the keywords "univariate", "time series", "nonstationarity", "transformation", "model", and "prediction". The query was executed on the Scopus database on September 2018 and returned 269 references.

Approximately 72 publications were selected to base the discussed review due to their relevance to the researched subject and impact (citation count). We have also performed an additional snowballing search in Google Scholar mainly on the subject of wavelet transform for time series prediction due to its increasing importance to the current research scenario on nonstationarity. This resulted in the addition of 6 papers to base our presented review. The selected 78 publications are presented in the timeline of Table 1.

The publications are presented in Table 1 in a chronological way, to provide a better visualization of the evolution of the adopted methods for addressing nonstationarity in univariate time series. The methods adopted or approached by each publication are represented within parenthesis after each authors names. Each publication is classified by the main tasks performed by their adopted methods.

According to the categorization in Fig. 2, we represent with different colors the publications which adopted mapping-based transformation methods, splitting-based transformation methods or that focus on other modeling techniques for treating nonstationarity. We also group the publications by their domain of application of the adopted methods, namely the statistical/natural sciences, socioeconomic/financial and industrial/business domains.

7.1. Domains of application

The timeline in Table 1 present 30 publications with a statistical/natural sciences domain of application, which mainly comprehend the problems of time series prediction and/or statistical modeling [45–57,26,58–60,10,61,62,21,63,64,34,22,65,66,37,67,68]. The works of Hipel [45] and Chilès and Delfiner [64], particularly, focus on problems regarding Geophysics and Geostatistics, respectively. Song et al. [37] perform analysis of geographic spatiotemporal series. Maier and Dandy [48], Otok and Suhartono [60], Pai and Ravishanker [62] and Nury et al. [66] all work with environment variables. In particular, Otok and Suhartono [60] aimed at modeling rainfall in Indonesia, and Nury et al. [66] focused on predicting future temperature measurements in northeastern Bangladesh.

We can also observe 32 publications with a socioeconomic [69,18,70,3,71,72] and financial [73,74,25,75,43,13,32,23,76–79] domain of application. Joo and Kim [30] conduct experiments using datasets with different origins including not only the financial market but also sales and airport demand, among others. The researches of Milionis [80], Lahmiri [29] and Corona et al. [81] handle bill, interest and inflation rates, respectively, while Gil-Alana and Jiang [82] evaluates the purchasing power parity hypothesis in China–United States relations. The authors [83] and [84] approach other socioeconomic issues such as tourism demand and security. Furthermore, the papers of Brandao and Nova [85], James and Murthy [86], Gao et al. [87], Wang et al. [35], Xiong et al. [88], Barba and Rodríguez [89] and Zhang et al. [90] present applications regarding traffic analysis.

Finally, we mention other 16 publications, with an industrial/business domain of application, which mainly refer to problems related to the energy market [91,92,33,31,44,24,15,41,14,36,93,94]. Among these publications, we note that Conejo et al. [33], Girish and Tiwari [24], Dudek [14], and Jamalmanesh et al. [94] focus on the electricity market, whereas Stefanakos and Schinas [44] and Akpinar and Yumusak [41] focus on the markets of marine fuel, and natural gas, respectively. An et al. [31] and Bokde et al. [95] study wind farm power and speed, respectively. Special cases are presented by Abraham and Balakrishna [96], Shu et al. [27] and Li et al. [97] which handle cargo transport, access loads and other time series problems relating industry and business.

7.2. Discussion on research evolution

From the timeline diagram in Table 1, we can observe that many of the applications presented by the selected publications fall into the statistical/natural sciences domain. This fact helps to suggest that the methods for handling nonstationarity in time series may still be widely found in statistical/theoretical researches. In that

Table 1
Timeline table of publications presenting some of the most researched methods for coping with nonstationarity in univariate time series. The publications are grouped by their domain of application and categorized by the main task of the applied methods (referenced within parenthesis).

Year	Statistical/Natural Sciences	Socioeconomic/Financial	Industrial/Business
1981	Hipel [45] (BCT, ARIMA, SARIMA) Stensholt and Tjøstheim [46] (ARIMA)		
1987			Sarma et al. [47] (DIF)
1992			Bhattacharya and Basu [48] (DIF)
1996	Baillie [49] (GARCH, FDIF, ARFIMA)		
2001	Maier and Dandy [50] (DT, DIF)		
2002	Dittmann and Granger [51] (FDIF)		
2003	D'Elia and Piccolo [52] (BCT, ARFIMA)		Abraham and Balakrishna [53] (GARCH, FDIF, ARIMA)
2004		Gil-Alana [54] (FDIF, ARFIMA) Milionis [55] (ARIMA)	
2005	Omtzigt and Paruolo [56] (DIF) Gil-Alana [57] (FDIF)		Conejo et al. [33] (ARIMA, WT)
2006	Ko and Vannucci [58] (ARFIMA, WT) Ko and Vannucci [60] (ARFIMA, WT) Fryzlewicz et al. [61] (WT)	Fryzlewicz and Nason [59] (GARCH, WT) Los [25] (GARCH, FT, WT) Hendry [62] (DIF) Marrocu [18] (LT, BCT, MAS, DT, FDIF, SDIF) Gospodinov et al. [63] (GARCH, FDIF, ARFIMA)	
2007	Haldrup and Nielsen [64] (FDIF) Brockwell [66] (ARFIMA) Morana [26] (FT) Palma [67] (MAS, DT, SM, GARCH, ARFIMA, WT)	Caporale and Gil-Alana [65] (FDIF)	
2008		Nachane and Clavel [13] (GARCH, FNLARMA, WT) Mills and Markellos [43] (DT, SM, GARCH, FDIF, ARIMA, SARIMA)	
2009	Cavaliere and Taylor [68] (GARCH) Otok and Suhartono [70] (ARIMA, SARIMA)	Brandao and Nova [69] (ARIMA, SARIMA)	
2010	Yang and Zurbenko [10] (LT, BCT, DT, SM, DIF, FT, KZFT, KZA, KZF) Gourieroux and Jasiak [72] (FDIF) Pai and Ravishanker [73] (ARFIMA, FDIF)	Minu et al. [71] (GARCH, WT) Ogasawara et al. [3] (MAS)	
2011	Jara [21] (FDIF)	Roshan et al. [32] (WT)	An et al. [31] (WT)
2012	Percival and Mondal [74] (WT) Chilès and Delfiner [76] (SM) Atto and Berthoumieu [34] (WPT)	James and Murthy [75] (DIF)	
2013	Maynard et al. [22] (FDIF)	Gao et al. [77] (WT) Gil-Alana and Jiang [78] (FDIF)	
2014	Ljung et al. [79] (ARIMA)	Claveria and Torra [80] (ARIMA)	Stefanakos and Schinas [44] (LT, BCT, DT) Shu et al. [27] (ARIMA, FT)
2015		Joo and Kim [30] (ARIMA, SARIMA, WT)	
2016		Wang et al. [35] (HW, ARIMA, EMD) Sadaei et al. [23] (ARFIMA) Lahmiri [29] (WT, EMD, VMD)	Sun et al. [36] (VMD) Girish and Tiwari [24] (THW, ETS, TF, ARIMA, ARFIMA) Chiroma et al. [15] (WT) Dudek [14] (PM) Akpinar and Yumusak [41] (HW, ARIMA)
2017	Nury et al. [81] (ARIMA, WT) Song et al. [37] (HHT) Douc et al. [85] (GARCH)	Corona et al. [82] (DIF) Corona et al. [84] (DIF) Zhang et al. [86] (EMD, ARIMA) Xiong et al. [87] (EMD) Rodriguez [88] (GARCH, FDIF) Ismail and Awajan [89] (EMD, ETS, ARIMA, SM, TF, HW) Barba and Rodríguez [90] (WT)	Liu et al. [83] (ARIMA, BCT, HW)
2018	Xie et al. [91] (ARIMA,DIF)	Zhang et al. [92] (WT, SARIMA) Ngene et al. [94] (FDIF) Caporale and Skare [96] (ARFIMA, GARCH, FDIF)	Bokde et al. [93] (EMD, DIF) Li et al. [95] (WT, ARIMA) Jamalmanesh et al. [97] (ARIMA)

Mapping-based transformations only Splitting-based transformations only Both mapping and splitting-based transformations Other techniques only

case, it is important to remark the demand for applying adequate methods for coping with nonstationarity in other real-world datasets, generally used in problems related to socioeconomics, financial market, and above all, industry. Fortunately, this seems to be the tendency since 2013, where most publications focus on real and non-theoretic applications.

Moreover, the analysis of Table 1 furthers the discussion of an increasing tendency regarding the use of splitting-based time series transformation methods. Although the timeline of the selected papers starts in 1981, the papers that apply methods that perform time series decomposition were published only since the year 2005. Furthermore, several of the papers that actually handle time series decomposition methods still apply them together with the more established mapping-based transformation methods. The amount of selected papers that focus solely on splitting-based transformations seems to increase since 2006. The highest concentration of such papers was published by the year 2016.

We may extend the observed late tendency to the use of nonparametric methods, which practically compose the category of splitting-based transformation methods. Since the growing tendency of works based on time series decomposition and nonparametric methods is relatively recent, it should suggest the need for further research on both subjects [13,10,15]. Table 2 presents a synthesis of the selected papers based on the timeline diagram in Table 1.

Also, we highlight the use of WT among the time series decomposition and nonparametric methods that have been most approached in recent years for business applications in time series prediction [29], having drawn unprecedented interest in its combination with machine learning methods to improve time series prediction accuracy [15]. In fact, 62.5% (20/32) of the selected papers in Table 1 that apply time series decomposition methods approach the use of the WT in an experimental application. Also, we note that 25% (8/32) of such papers are based on the use

Table 2
Synthesis of the selected publications presenting some of the most researched methods for coping with nonstationarity in univariate time series.

Domain of application	Publication	Applied transformations		
		Mapping-based only	Splitting-based only	Both
Statistical/Natural sciences	1981–2005	100%	0%	0%
	2006–2018	31%	25%	31%
Socioeconomic/Financial	1981–2005	100%	0%	0%
	2006–2018	53%	30%	17%
Industrial/Business	1981–2005	75%	0%	25%
	2006–2018	42%	33%	25%
All	1981–2005	93%	0%	7%
	2006–2018	47%	28%	20%

of another prominent decomposition method, namely the EMD, which has been increasingly adopted since 2016.

8. Experimental review

The research on the various nonstationary time series transformation methods reviewed in this paper pointed to the relevancy of an experimental comparison of their practical effects in the time series prediction problem. Such comparison may shed light on the advantages and limitations of these methods in practical applications. It can help researchers analyze their best options for treating nonstationarity. This paper fills this demand by devising and conducting a comparative analysis of eleven of the reviewed transformation methods that are most commonly used in practical applications. We applied these methods in the prediction of time series of five different datasets originated from time series prediction competitions and real macroeconomic observations collected by a government institution. The datasets used in this experimental evaluation were made available.[1] The next subsections describe the performed experiment in detail and discuss its results.

8.1. Datasets

Among the five time series datasets used in this experiment, three are benchmarks from time series prediction competitions (CATS [98], NN3 [99], and NN5 [100]). The other two datasets are provided by the Institute of Applied Economic Research of Brazil (Ipea) [101] and are derived from real economic and financial data of the world. When selecting these datasets, we were looking for a reasonable number of representative time series presenting different types of nonstationarity and statistical properties to provide a discussion on the effects of the evaluated transformation methods applied to the prediction of a diverse range of time series.

Moreover, this choice of datasets was made so as to encompass all domains of application of the publications reviewed in Section 7. Particularly, the CATS dataset represents the statistical/natural sciences domain, the NN3 and NN5 datasets represent the industrial/business domain, and the datasets provided by Ipea represent the socioeconomic/financial domain.

The CATS Competition dataset presents an artificial time series with 5000 observations, among which 100 are unknown. The unknown observations are grouped into five non-consecutive gaps of 20 successive values. The prediction of each gap may be considered a different problem, and each subset of the series followed by a gap may be considered a different time series to be modeled. In this context, we consider the CATS dataset as being composed of five time series of 980 observations. Both the NN3 and the NN5 Competition datasets present 111 time series. The series from the NN3 dataset have from 50 to 126 monthly observations drawn from a homogeneous population of real empirical business time series. All series from the NN5 dataset have 735 observations originated from daily withdrawals at 111 different cash machines within England, and may present missing data.

[1] https://github.com/RebeccaSalles/TSPred/wiki/nonstationary-review.

Table 3
Statistical tests results and analysis.

Statistical tests	H0	CATS	NN3	NN5	Ipea_M	Ipea_D
Breusch–Godfrey	Uncorrelated residuals	0%	37%	0%	0%	0%
Box–Pierce	Randomness	0%	30%	0%	0%	0%
Goldfeld–Quandt	Homoscedasticity	40%	91%	48%	70%	50%
White neural network	Linearity in mean	100%	81%	59%	83%	75%
ADF[a]	Nonstationarity	100%	62%	5%	100%	58%
KPSS[b]	Trend stationarity	0%	70%	71%	4%	8%
KPSS[b]	Level stationarity	0%	57%	50%	4%	0%

[a]Augmented Dickey–Fuller.
[b]Kwiatkowski–Phillips–Schmidt–Shin.

The two time series datasets provided by Ipea were selected as the most requested series collected in monthly and daily rates, and are henceforth referenced as Ipea_M and Ipea_D datasets, respectively. The Ipea is a public institution of Brazil that provides support to the federal government concerning public policies: fiscal, social, and economic. The data collected by Ipea and used in this experiment comprehend information on exchange rates (R$/US$), exports/imports prices, interest rates, minimum wage, unemployment rate, and more, measured from 1930 to September of 2017. Ipea_M contains 23 time series of 156 to 1019 observations. Ipea_D contains 12 time series of 901 to 8154 observations.

In order to obtain a better understanding of the statistical properties of the selected time series datasets, we have performed 7 of the most common statistical tests for autocorrelation, randomness and independence, heteroscedasticity, linearity, and stationarity. Table 3 contains a summary of the results of the statistical tests. For each dataset, it is presented the percentage of time series that had the null hypothesis test (H_0) confirmed.

The results in Table 3 show the considerable disparity in the statistical properties of the time series in the selected datasets. It is possible to observe that most time series did not confirm the null hypothesis of uncorrelated residuals and randomness. The heteroscedastic and linearity tests indicate that a substantial number of the time series present the properties of homoscedasticity and linear behavior around the mean. This result means that we can expect relative stability in the variance of the time series. Finally, the stationarity tests results show that a considerable amount of the time series over all datasets is nonstationary presenting unit root (also known as difference stationary), trend stationary (stationary around a deterministic trend) or level stationary (stationary around a level that changes over time). These latter results are particularly favorable for motivating the application of transformation methods such as the previously described in this paper.

8.2. Experimental settings

Several predictions were performed in this experiment using the presented datasets. The prediction horizons used in these predictions were set according to each dataset. The CATS, NN3 and NN5 competitions recommend a prediction horizon for their datasets which were adopted in this experiments. We predicted 20 observations ahead of the CATS times series (corresponding to the gaps of unknown values), 18 months ahead of the NN3 time series, and 56 days ahead of the NN5 time series. For the Ipea datasets, we set 12 months (a year) and 30 days (a month) as the prediction horizons for the Ipea_M and the Ipea_D datasets, respectively.

All predictions conducted in this experiment were performed using the ARMA models, defined as in Eq. (14) (fixing $d = 0$). The

Table 4
Nonstationary time series transformation methods selected for the experiment and the implemented data transformation process they apply.

Category	Transformation	Method	Reference
Mapping-based	Math. transform	Logarithmic transform	LT
	Math. transform	Logarithmic transform (base 10)	LT10
	Math. transform	Box–Cox transform	BCT
	Math. transform	Percentage changes transform	PCT
	Math. transform	Moving average smoother	MAS
	Detrending	Detrending	DT
	Differencing	Simple differencing	DIF
	Differencing	Seasonal differencing	SDIF
	Differencing	Simple and seasonal differencing	DIFs
Splitting-based	Decomposition	Wavelet transform	WT
	Decomposition	Empirical mode decomposition	EMD
None	None	None	Naive

use of ARMA models for prediction was combined with different preprocessing techniques performed by the nonstationary time series transformation methods listed in the next Section 8.3. The ARMA model is a well-established linear model, and it is often applied as a benchmark method for time series prediction [102]. We justify the choice of ARMA models for two reasons: (i) for providing a fair comparison of our generated predictions and (ii) for allowing the focus of this experiment discussion to be on the comparison of different nonstationary time series transformation methods (and their effects on prediction) rather than on modeling performances.

All the experiments produced several model fitting criteria (Akaike Information Criterion (AIC), corrected Akaike Information Criterion (AICc), Bayesian Information Criterion (BIC), and log-likelihood) and prediction accuracy measures (Mean Square Error (MSE), Normalized MSE (NMSE), Mean Absolute Percentage Error (MAPE), symmetric MAPE (sMAPE), and maximal error [103]). Nevertheless, we henceforth focus on the MSE measurements to simplify the discussion of the prediction results, since it is generally the most commonly used measure for assessing prediction accuracy.

8.3. Implementation of nonstationary time series transformation methods

We considered important to explore transformation methods of different categories of the presented review in this comparative analysis to provide a complete overview of the effects of the reviewed methods on prediction. Thus, among the reviewed nonstationary time series transformation methods we opted for comparing the eleven most commonly used in practical applications. For comparison purposes, we also included in the experimental evaluation the naive approach of not applying any data transformation before prediction. Table 4 presents the selected transformation methods (and their reference acronyms) separated by the implemented data transformation process they apply and by their respective major categories.

All selected transformation methods were implemented in the R-package *TSPred* [104]. It was originally developed for the problem of benchmarking time series prediction using automatically fitted linear models. The new version 4.0 of *TSPred* has been updated with the implementation of the selected reviewed nonstationary time series transformation methods. It includes functions for automatically finding optimized parameters and for analyzing prediction results using different nonstationary time series methods.

The transformation methods LT, LT10, BCT, PCT, and MAS were implemented as mathematical transforms following their characteristic equations. They were applied to the time series observations before prediction. For the MAS method, we implemented a function for optimizing its order of moving average according to the best predictions from a cross-validation dataset. The transformation method DT was implemented and provides a fittest polynomial regression (PR) model for detrending the original time series before prediction.

All transformation methods referenced as DIF, SDIF, and DIFs perform the task of differencing a time series before modeling and predicting. The DIF, SDIF, and DIFs respectively correspond to the simple, seasonal, and combined simple/seasonal differencing processes. These methods are available within the ARIMA modeling process of the *auto.arima* function of the *forecast* R-package [105]. It performs KPSS tests to determine their differencing parameters.

Both EMD and WT perform the process of time series decomposition before the prediction. For the EMD, we used functions from the R-package *EMD* [106], while for the WT we relied on the *wavelets* R-package [107]. Analogously to the MAS method, we implemented functions for optimizing the parameters of EMD (meaningful IMFs and boundary condition) and WT (filter and level of decomposition) according to the best predictions in a cross-validation dataset. Particularly for the WT, we implemented the MODWT transform. The decomposed time series were predicted separately by ARMA models in the case of WT. In the case of EMD, the IMFs were predicted by a vector autoregressive (VAR) model and the residue component (residue signal after extracting IMFs) was predicted by the fittest PR model. This latter definition is due to the implementation requests of the *EMD* package.

Regarding these previously discussed transformation methods, all predictions for the original time series of the selected datasets are obtained using inverse transforming of the predicted series. Particularly, for the method referenced as Naive, it implies that the naive approach was adopted and no data transformation was performed. In this case, the time series is directly predicted by an ARMA model. We also note that all linear models used in this experiment, namely, the ARMA, ARIMA, and PR were implemented by functions that automatically optimize their parameters. The parameter optimization is done either by performing prediction tests with cross-validation (as it is the case for PR) or by conducting the Hyndman and Khandakar algorithm [108] (the case for ARIMA/ARMA models). All functions used in this experiment are available in the version 4.0 of the R-package *TSPred*. For more details on the implementation of the transformation methods in this experiment, please refer to the documentation of *TSPred* [104]. All code, experimental data, and results are made available (see footnote 1).

8.4. Results and discussion

In this section, we discuss the results obtained with our comparative experiments for the selected nonstationary time series transformation methods. For that, we have used the MSE error measures produced by the predictions. The next subsections present our analysis by focusing on the results generated over a single time series (Section 8.4.1) and over all time series of each selected dataset (Sections 8.4.2 and 8.4.3). We also discuss results across all datasets (Section 8.4.4) and the observed impact caused by transformation methods on prediction accuracy (Section 8.4.5).

8.4.1. Analysis over a single time series

We have produced Taylor diagrams [109] for summarizing in a single plot the different aspects of the effects of each transformation method in the prediction of a particular time series. We analyzed two time series of the CATS dataset, as it is one of the most nonstationary datasets among the selected. The diagrams are depicted in Fig. 3.

Figs. 3a and 3b present prediction results for the time series corresponding to the second and third sequences of known values

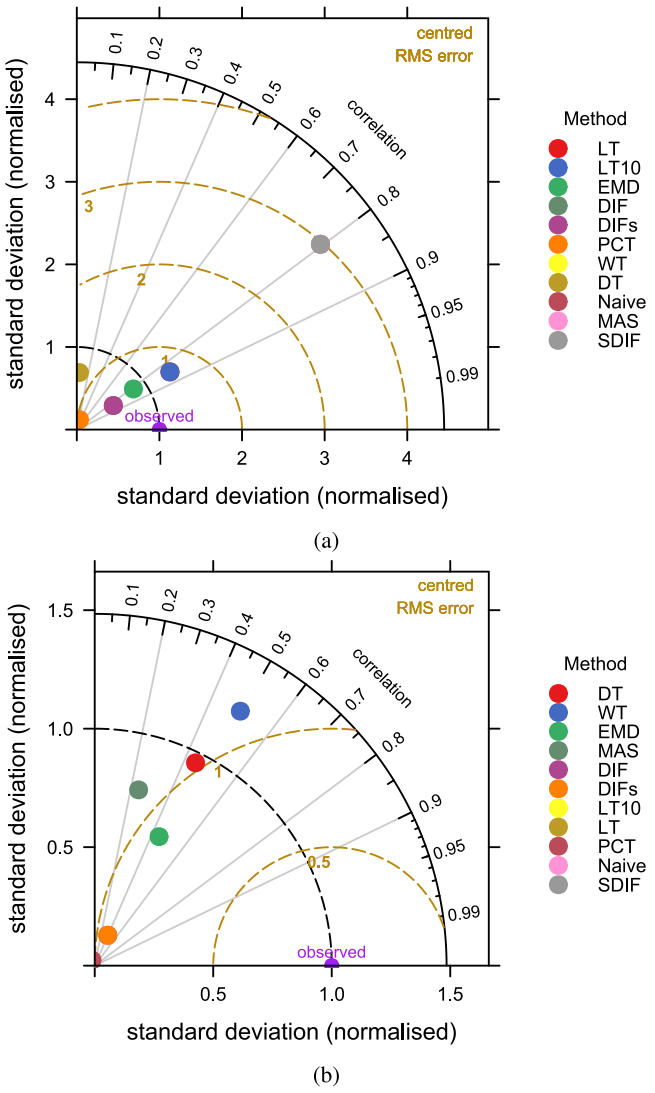


Fig. 3. Taylor diagrams of predictions for two CATS time series preprocessed by different nonstationary time series transformation methods. (a) presents results for the second sequence of known values 1001–1980 of CATS and (b) presents results for the third sequence of known values 2001–2980 of CATS.

(1001–1980 and 2001–2980) of the CATS times series. Taylor diagrams illustrate the quality of predictions against reference values, which in this case are the real observations of the time series (that were to be predicted). Data regarding the real observations of the time series, as well as the predictions corresponding to the use of each nonstationary time series transformation method, are represented in the diagram by points inside a quarter circle, where the former is labeled as "observed". The sections of the quarter circle correspond to the level of correlation between the predicted and observed values. The dashed brown contours centered in the observed point represent the level of root mean square errors (RMSE) of the predictions. The distances to the origin correspond to the standard deviation of the predictions, and the dashed black contour centered in the origin mark the standard deviation of the observed values. Small distances from the observed point mean better predictions. Particularly for our plots in Fig. 3, the legends also present the methods ranked by smallest MSE errors.

The predictions in Fig. 3a present higher correlation and closer standard deviations to the observed values, besides smallest RMSE errors. Furthermore, the distances to the observed point seem to produce a similar ranking to the one in the legend, based on the MSE errors. Conversely, the prediction results in Fig. 3b make the identification of the best transformation method (for simplicity, best method) less clear. Although the results from EMD are closer to the observed, the results of DT present a more similar standard deviation, and the results of WT a higher level of correlation. Moreover, the distances to the observed point suggest a different ranking than the one in the legend, even when only the RMSE error is considered. That indicates the need for special attention when selecting the error measure to be used during analysis since the assessment of the best method was not consistent for both measures.

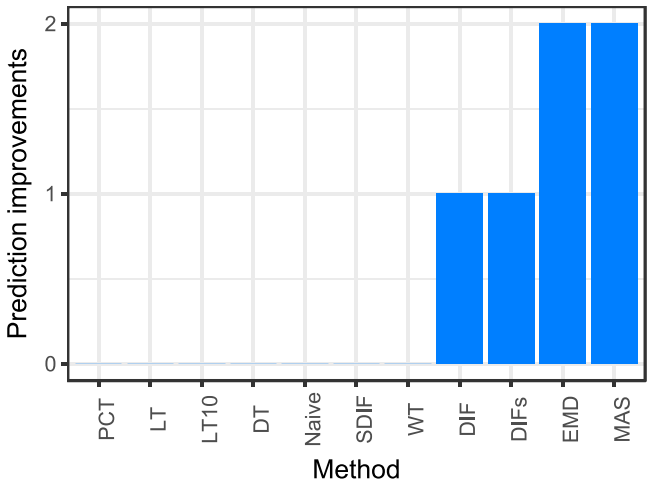
8.4.2. Analysis over a single dataset

We also considered important to summarize prediction results over all the time series in a particular dataset. For that, we have performed pairwise Wilcoxon statistical tests for comparing the MSE errors yielded from predictions corresponding to all nonstationary time series transformation methods evaluated against each other. A visualization of the number of statistically significant prediction improvements provided by the use of each nonstationary times series transformation method over others in the experiments for the CATS dataset is presented in Fig. 4a. Based on this plot, we may consider DIF, DIFs, EMD, and MAS among the best methods. The use of the Naive approach and the SDIF transformation method did not provide statistically significant prediction improvements over other methods. This was an expected result since all time series in the CATS dataset are nonstationary. The use of SDIF derived prediction performance (for short, performance) equal to the Naive method since no seasonal parameters were estimated by the Hyndman and Khandakar algorithm [108], suggesting a low level of seasonality in the series. Also, the DIF and DIFs methods both represent the method DIF for the same reason.

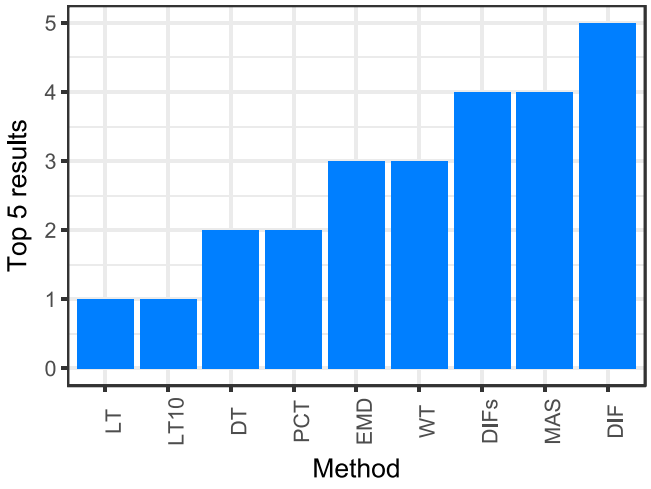
Fig. 4 also presents another indicator that may be used for assessing the quality of the predictions. Namely, the number of times the use of each nonstationary times series transformation method yielded one of the five smallest absolute values of MSE prediction errors across all time series. That is, the number of times the use of each transformation method was responsible for one of the top 5 results of the time series in CATS. This information is presented in Fig. 4b.

The time series of the CATS dataset presented considerable nonstationarity in the form of difference stationarity, which motivates the application of stochastic trend removal or differencing, which uses the operator in Eq. (13), to achieve stationarity. This fact favored the transformation method DIF, which produced the best predictions, followed by the DIFs, EMD and WT, as can be observed in Fig. 4b. The results of MAS were again among the best, even though there was no significant seasonality as confirmed by the Hyndman and Khandakar algorithm. This may indicate the presence of clear long-term trends in the series generated by Eq. (10).

Furthermore, the CATS time series presented no trend or level stationarity, which indicates that the removal of a deterministic trend may not be enough for providing stationarity. This fact helps explain the results obtained by the use of DT. Another important property of the CATS time series is that they presented the largest rate of heteroscedasticity (60%) among all datasets. The heteroscedasticity in the series possibly favored EMD and WT that are prepared to handle non-constant variability. It is also true for the PCT, LT, LT10, and BCT methods. The application of these latter transformation methods, however, was impaired due to the presence of negative data in all CATS time series, which caused the PCT, LT, and LT10 to produce NA values. Due to that, necessary parameters could not be computed for BCT in any time series, therefore not being present in the plots. This behavior is consistent with the Eqs. (7)–(9). Finally, we highlight the performances yielded by the use of both EMD and WT as the splitting-based time series decomposition methods evaluated in this experiment.



(a)



(b)

Fig. 4. Plots of the results for the prediction of all CATS time series preprocessed by each selected nonstationary time series transformation methods. (a) presents the number of statistically significant prediction improvements provided by the use of each nonstationary times series transformation method over others. (b) presents the number of times the use of each transformation method was responsible for one of the top 5 prediction results of the time series.

8.4.3. Analysis of results on each selected dataset

Next, we present the results of the experiments for each selected dataset in Fig. 5. The presented scatter plots summarize the same indicators presented in Fig. 4 for all CATS, NN3, NN5, Ipea_M, and Ipea_D datasets. That is, they summarize (i) the number of times the use of each nonstationary time series transformation method provided a statistically significant prediction improvement over others and (ii) the number of times the use of each method was responsible for one of the top 5 prediction results of the time series (based on the MSE prediction errors). By analysis of the results depicted in Fig. 5, we defined that the use of a transformation method produced better predictions if (i) it was responsible for one of the top 5 results for the majority of time series in a dataset, or (ii) it provided at least half of the maximal number of statistically significant improvements in a dataset. Conversely, the transformation methods that led to worse predictions were highlighted in Fig. 5 with a dashed rectangle. As the results regarding the CATS dataset were previously examined, we henceforth focus our discussion on the remaining datasets.

Results on NN3

The results for the NN3 dataset are presented in Fig. 5b. Most time series of the NN3 dataset (62%) were difference stationary, also favoring the transformation methods DIF and DIFs. It was similar to the results for the CATS dataset. Particularly, DIF and DIFs provided statistically significant prediction improvements over eight of the twelve evaluated transformation methods. However, surprisingly, the results produced by the use of the Naive approach were among the best despite nonstationarity. This result may be due to a combination of properties presented by the series such as the high level of linearity (81%), the lowest rate of heteroscedasticity (9%) among the datasets and a considerable amount of autocorrelation, which are all beneficial to ARMA models. These same properties may also have contributed to the somewhat unsatisfactory results of EMD since it tends not to perform well when data presents similar frequencies. Also, since EMD is a flexible nonparametric method, overfitting is a possibility. Furthermore, despite the NN3 time series presenting 70% trend and 57% level stationarity, DT, which removes a deterministic trend by Eq. (11), was not among the best methods, suggesting that a polynomial regression as η_t may not provide an explanation of the series behavior. Also, performances from LT10, PCT, LT, and BCT were considered among the best. They were favored by strictly positive data and possible stability in the relative percentage of change in observations. Other interesting results were obtained by the use of MAS, SDIF, and WT transformation methods. Although being able to produce reasonable results, they were still not among the best methods, probably due to the presence of unclear seasonality in the data, in the case of MAS and SDIF, and possible overfitting by the flexible representation of Eq. (16), in the case of WT, since the data was mostly linear.

Results on NN5

For an overview of the results of the NN5 dataset, Fig. 5c may be observed. The dataset NN5 time series presented the lowest rate of difference stationarity among all datasets. This fact particularly favored the Naive approach, as could be expected, and it was among the best methods. Also, since the NN5 time series presented 71% trend and 50% level stationarity, the deterministic trend removed by DT was suitable for providing stationarity. Other properties presented by the NN5 series are heteroscedasticity in 52% of the series and nonlinearity in 41% of the series (highest rate among the datasets). Transformation methods which are prepared to handle such properties may have been favored, such as WT. Other examples of such methods, like PCT, LT10, LT, BCT, and EMD could have also been favored, but that was not the case, due to 0 values in 93 of the 111 series of the dataset (impairing the PCT, LT10, LT, and BCT methods) and possibly due to the presence of similar frequencies in the EMD decomposed data, causing worse performances. Conversely, the transformation methods DIF, SDIF, and DIFs produced better predictions. We highlight the results of MAS being the best method, which indicates the presence of clear long-term behaviors and seasonality in the series, which is to be expected in daily withdrawals at cash machines.

Results on Ipea_M

For discussing the results of the Ipea_M dataset we use Fig. 5d. All series in Ipea_M were actually difference stationary. This property expectedly favored transformation methods like DIF, DIFs, EMD, and WT that remove stochastic trends, apply differencing or use decomposition to derive a stationary time series before prediction. It also impaired the Naive approach since it does not address this property. As it was for the CATS results, SDIF presented performance equal to Naive as no seasonal differencing parameters were estimated by the *auto.arima* function [108]. In spite of having trend and level stationarity on only less than 5% of the time series,

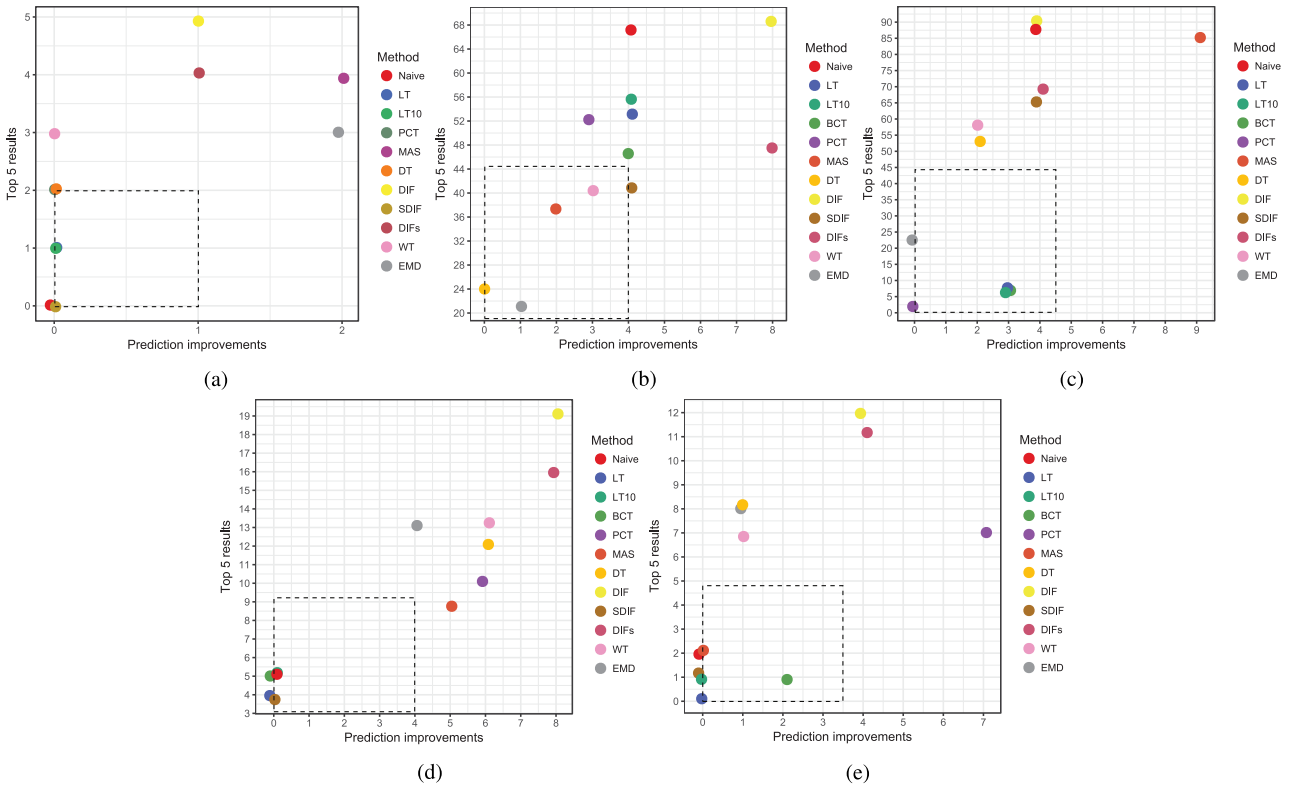


Fig. 5. Scatter plots for summarizing the results indicators shown in Fig. 4 for all time series in the datasets (a) CATS, (b) NN3, (c) NN5, (d) Ipea_M and (e) Ipea_D. The dashed rectangles mark the transformation methods that led to worse predictions.

DT presented a reasonable performance, probably due to the help of mostly constant variability (only 30% of heteroscedasticity) and high level of linearity (83% linearity in the mean) presented by the time series in the dataset. Results of LT, LT10, and BCT were poor compared to other methods but PCT presented reasonable performance. Although all these latter methods were badly affected by the negative and/or zero values in 26% of the time series in the dataset, PCT benefit from the possible constancy in the percentage of change in the data of economic nature. The MAS method also presented fair performance, favored by the seasonality characteristic of the time series.

Results on Ipea_D

Finally, for the results of the Ipea_D dataset, we present Fig. 5e. Difference stationarity was present in 58% of the series in Ipea_D, impairing the Naive approach which does not address it, and favoring the transformation methods DIF, DIFs, WT, and EMD, which were among the best methods, followed by PCT and DT. The performance of DT was unexpected since there is only 8% of the series in the dataset that present trend or level stationarity, thus not suggesting the need for removal of a deterministic trend for achieving stationarity. The presence of 75% of linear and reasonable 50% of homoscedastic time series may have been advantageous for the performance provided by the use of DT. Non-constant variability in 50% of the series, together with a possible constancy in the percentage of change of the economic data, may have also been in favor of the performance obtained by the use of PCT, that was also among best methods presenting statistically significant prediction improvements over seven other methods. Despite being able to handle variability in the data, much like PCT, the transformation methods LT, LT10, and BCT did not perform well. Since all four transformation methods were not resilient to 20% of the series containing 0 values, a superiority of PCT over this dataset is indicated. Furthermore, we note that (for the same reason as presented in previously discussed datasets) SDIF represent the same method as Naive. Other results that are worth mentioning are the ones produced by the use of MAS, which was not among the best methods, probably due to the presence of unclear long-term trends or seasonality in the data.

8.4.4. Summary of results across all datasets

In order to have an overview of the predictions provided by the use of each of the evaluated nonstationary time series transformation methods over all the selected datasets, we generate the plot in Fig. 6 which presents the relative number of times that the use of each transformation method led to better predictions across all datasets. In other words, this plot presents the relative number of times each method was not present within any area demarked by a dashed rectangle in Fig. 5, which represent the worst evaluated transformation methods regarding the experiments on each dataset.

From the analysis of Fig. 6, we draw attention to the results of the methods DIF and DIFs, which were consistently among the best methods. This fact indicates the adequacy of differencing techniques for aiding the prediction of the time series of the selected datasets. The splitting-based time series decomposition methods EMD and WT were also among the best methods throughout the experiments, together with MAS, PCT, and DT. We also note that the use of WT provided a more consistent performance than EMD in our experimental analysis. The worst evaluated transformation methods across all datasets (marked before the red vertical line) were expectedly the Naive approach, and the SDIF, LT, LT10, and BCT transformation methods.

8.4.5. Discussion

Our experimental results indicate that the effects of each evaluated nonstationary time series transformation methods regarding predictions and coercion of a time series towards stationarity were

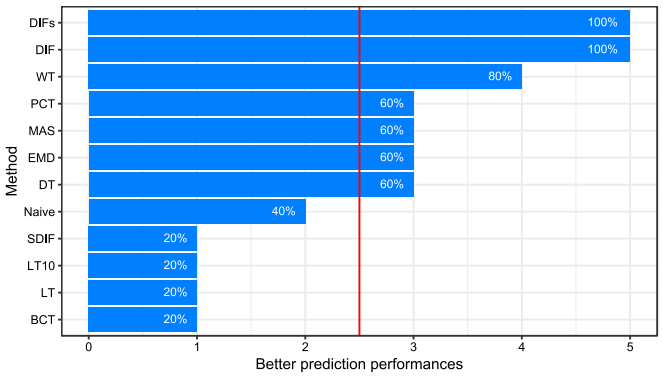


Fig. 6. Plot of the number of times (and percentage) the use of each method provided better predictions over all five evaluated datasets.

dependent on the statistical properties and characteristic nature of the data. This observation and the analysis of the information in Fig. 6 suggest the importance of a study on the effects and benefits of different nonstationary time series transformation methods to be performed on the dataset at hand before any prediction tasks.

Moreover, the significance of performing a preliminary data transformation activity may be better visualized in Fig. 7. The plots in Fig. 7 depict relevant information about observed prediction accuracy improvements (measured by the decrease in MSE error) provided by the best method over the Naive approach for each time series used in this experimental analysis.

As can be observed in Fig. 7a, the datasets for which transformation methods were most effective were CATS, Ipea_D and Ipea_M. This result may be mostly due to the fact that these are among the datasets which presented the greatest rates of difference stationarity in their time series. This is one of the most challenging forms of nonstationarity and therefore, when not treated, has a significant negative impact on the accuracy of simpler prediction methods such as the ARMA model.

Furthermore, the lower relative accuracy improvements observed for the NN3 and NN5 datasets are probably due to properties of linearity, homoscedasticity, autocorrelation, and also a low rate of difference stationarity (in the case of the NN5 dataset), which are favorable features to the performance of ARMA models. By analyzing the last box plot of Fig. 7a, corresponding to all time series of all datasets, we observe a skew in the results where the best methods lead to prediction accuracy improvements of at most around 30% in half of the time series. This skew is actually to be expected since most data are from the NN3 and NN5 datasets that provided 222 out of the 262 evaluated time series. Nonetheless, the upper quartile and the upper extreme values of the last box plot represent considerably high prediction accuracy improvements obtained by transformation methods over half of the evaluated time series.

Complementary information may be obtained from Fig. 7b that illustrates the number of evaluated time series across all datasets for which at least a minimum percentage of prediction accuracy improvement was achieved by the use of their best method. The plot in Fig. 7b confirms and detail the information presented in the last box plot of Fig. 7a, showing that for almost 50% of the series the best methods produced prediction accuracy improvements of at least 30%. Moreover, we can observe other significantly higher rates of accuracy improvements reaching at least 70% for 20% of the series, at least 95% for over 10% of the series and a maximum value of at least 99% prediction accuracy improvement for 7% of the time series used in our experiments. The results indicate the potential impact of adequate transformation methods on the problem of accurately predicting times series presenting nonstationary properties.

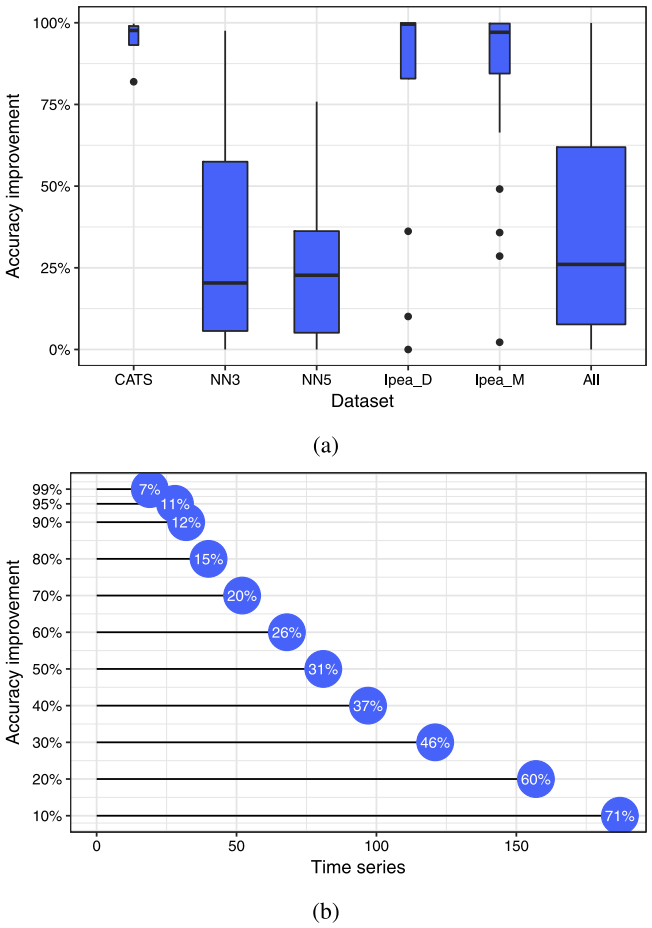


Fig. 7. Plots of prediction accuracy improvement provided by transformation methods. (a) presents box plots of prediction accuracy improvements provided by the best method for each time series of each dataset. (b) presents the number (and percentage) of time series (over all datasets) for which at least a minimum percentage of prediction accuracy improvement was provided by their best method.

However, the task of selecting an adequate transformation method for the problem of predicting a particular nonstationary time series is not straightforward. This is confirmed by our previous analysis of the effects of the evaluated nonstationary time series transformation methods on prediction. In particular, our experimental results support a preliminary conclusion that there is no "silver bullet" method for all datasets. Also, the notion of an adequate transformation method is subjective according to each time series prediction application.

Our experiments pointed to the influence of transformation methods during prediction. Nonetheless, the question remains "is it possible to consistently find appropriate transformation methods for the prediction of a nonstationary time series before testing?". To address this question, we applied a validation phase during model training. The validation was introduced to rank candidate methods based on MSE. The suitability of the method selected during validation could than be confirmed during the testing phase of the prediction application [2] using a similar ranking process. Fig. 8 presents plots for comparing rankings of transformation methods based on MSE measures generated from predictions performed during both validation and testing phases. Analogously to our previous discussion, henceforth we consider the best method the one whose application resulted in the smallest absolute values of MSE prediction errors.

We may consider the rate at which the best method observed during the validation phase is found among the top 5 methods

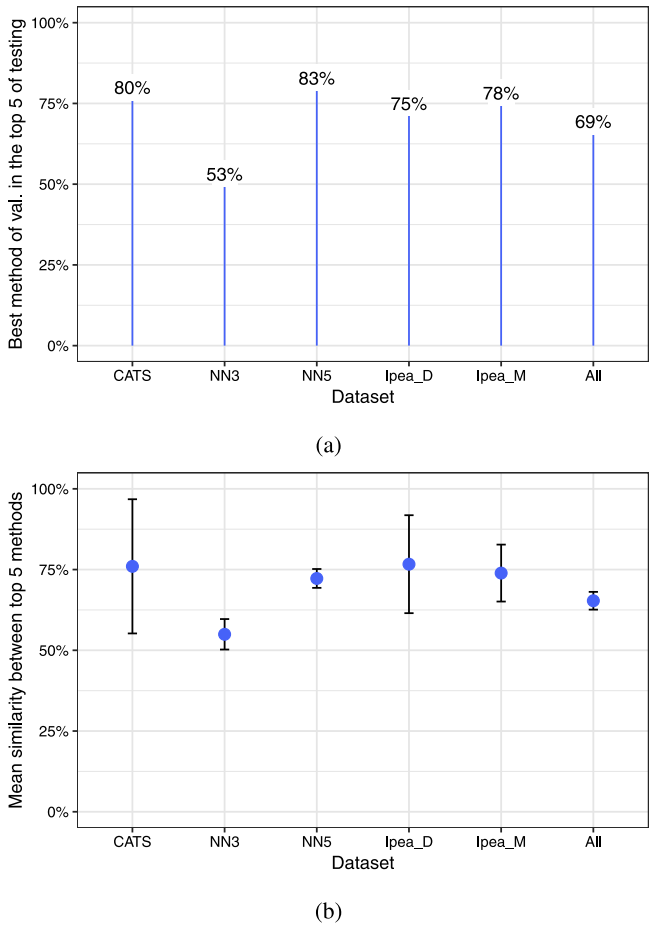


Fig. 8. Plots for comparing the best methods according to predictions performed during validation and testing phases. (a) presents the (percentage) number of times the best method found during validation was also present in the top 5 ranked methods found during testing, for each time series of each dataset. (b) presents the mean (percentage) similarity (intersect) between the top 5 ranked methods of both validation and testing phases, for each time series of each dataset.

found during the final testing phase. This information is presented in Fig. 8a where it is possible to see that, for the majority of the time series adopted in our experiments, the best method found during the validation was in fact confirmed as one of the most adequate (if not the most adequate) transformation methods for aiding the accurate prediction of each time series. We also draw attention to the lowest rate of 53% derived from the time series of the NN3 dataset. This rate is mostly due to the relatively small length of the NN3 times series (the smallest among the adopted datasets), a property that is generally challenging for time series modeling. This result helps to suggest that inferences on best methods made on validation phase are more likely to approximate the truth when sufficient data is available.

The plot in Fig. 8b presents another indicator for evaluating the ability of consistently finding adequate transformation methods for aiding the prediction of a time series through a validation phase. It depicts the mean rate of similarity (intersect) between the top 5 best methods ranked during the validation and testing phases. Interpretation of the plot suggests that even in case the best method found in validation phase is not present in the top 5 methods of the testing phase, the top 5 rankings of both phases share on average at least 65% (3) of their transformation methods considering all adopted time series. This observation denotes a relative consistency among the results of both the validation and testing phases. Confidence intervals are also represented. Expectedly, the three wider confidence intervals regarding CATS, Ipea_D and Ipea_M correspond precisely to the datasets which possess the lowest number of time series (respectively 5, 12 and 23 in comparison to 111 for each NN3 and NN5). The analysis of the results illustrated in Fig. 8 indicates that it is generally likely to find and select the most adequate nonstationary time series transformation methods for aiding the production of accurate predictions based on the use of a validation phase.

9. Final remarks

This paper presents a review of nonstationary time series transformation methods for time series prediction. A categorization of such transformation methods was presented together with a timeline obtained through a systematic mapping study. Furthermore, the paper presents a comparative experimental analysis and discusses the effects of some of the reviewed transformation methods on the problem of time series prediction. For this analysis, we selected eleven methods of the most commonly used in practical applications. The aim of this paper is contributing to the process of analysis, selection, and application of nonstationary time series transformation methods.

An overview of the effects of the evaluated methods regarding predictions and stationarity was produced based on our experimental results. Although it was possible to note a somewhat consistency in the results of the evaluated transformation methods, there was no uniquely best method across all datasets, and the nature and statistical properties of the time series were especially relevant to the results.

Nonetheless, it was possible to observe better predictions when transformation methods based on differencing and moving average smoothing were applied before the prediction of the time series of the selected datasets. Transformation methods that perform time series decomposition, which have been an object of increasing attention, were also among the best methods. Among the worst methods was, as expected, the naive one, where no data transformation is performed before prediction. Particularly, this approach provided predictions with significantly lower accuracy when compared to the case in which nonstationarity was treated.

Additionally, results indicate that the use of a validation phase for exploring different transformation methods generally leads to the selection of one of the most appropriate for obtaining accurate time series predictions. Our experimental results suggest as future trends (i) the increase in the importance of the process of data transformation for the problem of accurate prediction of nonstationary time series and (ii) the need for studying and evaluating suitable methods to perform this activity according to the dataset at hand.

Acknowledgments

The authors thank CNPq, Brazil, CAPES, Brazil (Finance Code 001), and FAPERJ, Brazil for partially sponsoring this research.

References

[1] D. Pyle, Data Preparation for Data Mining, first ed., Morgan Kaufmann, San Francisco, Calif, 1999.
[2] J. Han, M. Kamber, J. Pei, Data Mining: Concepts and Techniques, third ed., Morgan Kaufmann, Haryana, India; Burlington, MA, 2011.
[3] E. Ogasawara, L.C. Martinez, D.d. Oliveira, G. Zimbrao, G.L. Pappa, M. Mattoso, Adaptive Normalization: A novel data normalization approach for non-stationary time series, in: The 2010 International Joint Conference on Neural Networks (IJCNN), 2010, pp. 1–8.
[4] P. Esling, C. Agon, Time-series data mining, ACM Comput. Surv. 45 (1) (2012) 1–34.

[5] K. Buza, Time series classification and its applications, in: Proceedings of the 8th International Conference on Web Intelligence, Mining and Semantics, in: WIMS '18, ACM, New York, NY, USA, 2018, pp. 4:1–4:4.
[6] C. Cheng, A. Sa-Ngasoongsong, O. Beyca, T. Le, H. Yang, Z. Kong, S. Bukkapatnam, Time series forecasting for nonlinear and non-stationary processes: A review and comparative study, IIE Trans. (Inst. Ind. Eng.) 47 (10) (2015) 1053–1071.
[7] D. Gujarati, Basic Econometrics, fourth ed., McGraw-Hill/Irwin, Boston; Montreal, 2002.
[8] R.H. Shumway, D.S. Stoffer, Time Series Analysis and Its Applications: With R Examples, fourth ed., Springer, New York, NY, 2017.
[9] R.S. Tsay, Analysis of Financial Time Series, third ed., Wiley, Cambridge, Mass, 2010.
[10] W.a. Yang, I.b. Zurbenko, Nonstationarity, Wiley Interdiscip. Rev. Comput. Stat. 2 (1) (2010) 107–115.
[11] D.M. Hanssens, L.J. Parsons, R.L. Schultz, Market Response Models: Econometric and Time Series Analysis, second ed., Springer, Boston, Mass., 2003.
[12] G.E.P. Box, G.M. Jenkins, G.C. Reinsel, Time Series Analysis: Forecasting and Control, fourth ed., Wiley, Hoboken, N.J, 2008.
[13] D. Nachane, J. Clavel, Forecasting interest rates: A comparative assessment of some second-generation nonlinear models, J. Appl. Stat. 35 (5) (2008) 493–514.
[14] G. Dudek, Neural networks for pattern-based short-term load forecasting: A comparative study, Neurocomputing 205 (2016) 64–74.
[15] H.a.h. Chiroma, S.a. Abdul-Kareem, A.b.S.M. Noor, A.I.c. Abubakar, N.d.S. Safa, L.e. Shuib, M.f.F. Hamza, A.g.Y. Gital, T.e. Herawan, A review on artificial intelligence methodologies for the forecasting of crude oil price, Intell. Autom. Soft Comput. 22 (3) (2016) 449–462.
[16] K.P. Murphy, Machine Learning: A Probabilistic Perspective, first ed., The MIT Press, Cambridge, MA, 2012.
[17] J.D. Cryer, K.S. Chan, Time Series Analysis: With Applications in R, second ed., Springer, New York, 2010.
[18] E. Marrocu, An investigation of the effects of data transformation on nonlinearity, Empir. Econ. 31 (4) (2006) 801–820.
[19] I.K. Yeo, R. Johnson, A new family of power transformations to improve normality or symmetry, Biometrika 87 (4) (2000) 954–959.
[20] Z. Wu, N. Huang, S. Long, C.K. Peng, On the trend, detrending, and variability of nonlinear and nonstationary time series, Proc. Natl. Acad. Sci. USA 104 (38) (2007) 14889–14894.
[21] E.C. Jara, Long memory time series forecasting by using genetic programming, Genet Programm. Evolvable Mach. 12 (4) (2011) 429–456.
[22] A. Maynard, A. Smallwood, M. Wohar, Long memory regressors and predictive testing: a two-stage rebalancing approach, Econometric Rev. 32 (3) (2013) 318–360.
[23] H. Sadaei, R. Enayatifar, F. Guimarães, M. Mahmud, Z. Alzamil, Combining ARFIMA models and fuzzy time series for the forecast of long memory time series, Neurocomputing 175 (2016) 782–796.
[24] G.P.a. Girish, A.K.b. Tiwari, A comparison of different univariate forecasting models for spot electricity price in India, Econ. Bull. 36 (2) (2016) 1039–1057.
[25] C. Los, Financial Market Risk: Measurement and Analysis, first ed., Routledge, London; New York, 2006.
[26] C. Morana, Multivariate modelling of long memory processes with common components, Comput. Statist. Data Anal. 52 (2) (2007) 919–934.
[27] M.-H. Shu, B.-M. Hsu, T.-L. Nguyen, C. Lu, J.C. Huang, Forecasting cargo throughput with modified seasonal ARIMA models, WSEAS Trans. Math. 13 (2014) 171–181.
[28] L. Xi, L. Kong, S. Xiao, G. Chen, Prediction model of non-stationary time series parameters for a complex blending process, in: 26th Chinese Control and Decision Conference, CCDC 2014, 2014, pp. 1027–1030.
[29] S. Lahmiri, Interest rate next-day variation prediction based on hybrid feedforward neural network, particle swarm optimization, and multiresolution techniques, Physica A 444 (2016) 388–396.
[30] T. Joo, S. Kim, Time series forecasting based on wavelet filtering, Expert Syst. Appl. 42 (8) (2015) 3868–3874.
[31] X. An, D. Jiang, C. Liu, M. Zhao, Wind farm power prediction based on wavelet decomposition and chaotic time series, Expert Syst. Appl. 38 (9) (2011) 11280–11285.
[32] W. Roshan, R. Gopura, A. Jayasekara, Financial forecasting based on artificial neural networks: Promising directions for modeling, in: 2011 6th International Conference on Industrial and Information Systems, ICIIS 2011 - Conference Proceedings, 2011, pp. 322–327.
[33] A.J. Conejo, M.A. Plazas, R. Espinola, A.B. Molina, Day-ahead electricity price forecasting using the wavelet transform and arima models, IEEE Trans. Power Syst. 20 (2) (2005) 1035–1042.
[34] A.M.a. Atto, Y.b. Berthoumieu, Wavelet packets of nonstationary random processes: Contributing factors for stationarity and decorrelation, IEEE Trans. Inform. Theory 58 (1) (2012) 317–330.
[35] H. Wang, L. Liu, S. Dong, Z. Qian, H. Wei, A novel work zone short-term vehicle-type specific traffic speed prediction model through the hybrid EMDARIMA framework, Transportmetrica B 4 (3) (2016) 159–186.
[36] G. Sun, T. Chen, Z. Wei, Y. Sun, H. Zang, S. Chen, A carbon price forecasting model based on variational mode decomposition and spiking neural networks, Energies 9 (1) (2016).
[37] W. Song, L. Wang, Y. Xiang, A. Zomaya, Geographic spatiotemporal big data correlation analysis via the HilbertHuang transformation, J. Comput. System Sci. 89 (2017) 130–141.
[38] M.P. Clements, D.F. Hendry (Eds.), A Companion to Economic Forecasting, 1 edition, Wiley-Blackwell, Malden, Mass., 2005.
[39] J. Fox, Applied Regression Analysis and Generalized Linear Models, third ed., SAGE Publications, Inc, Los Angeles, 2015.
[40] I. Moghram, S. Rahman, Analysis and evaluation of five short-term load forecasting techniques, IEEE Trans. Power Syst. 4 (4) (1989) 1484–1491.
[41] M. Akpinar, N. Yumusak, Year ahead demand forecast of city natural gas using seasonal time series methods, Energies 9 (9) (2016).
[42] R. Carmona, Statistical Analysis of Financial Data in R, second ed., Springer, Heidelberg, 2013.
[43] T. Mills, R. Markellos, The econometric modelling of financial time series, 2008.
[44] C. Stefanakos, O. Schinas, Forecasting bunker prices; a nonstationary, multivariate methodology, Transp. Res. C 38 (2014) 177–194.
[45] K.W. Hipel, Geophysical model discrimination using the akaike information criterion, IEEE Trans. Automat. Control 26 (2) (1981) 358–378.
[46] E. Stensholt, D. Tjøstheim, Factorizing multivariate time series operators, J. Multivariate Anal. 11 (2) (1981) 244–249.
[47] R.T. Baillie, Long memory processes and fractional integration in econometrics, J. Econometrics 73 (1) (1996) 5–59.
[48] H.R. Maier, G.C. Dandy, Neural network based modelling of environmental variables: A systematic approach, Math. Comput. Modelling 33 (67) (2001) 669–682.
[49] I. Dittmann, C. Granger, Properties of nonlinear transformations of fractionally integrated processes, J. Econometrics 110 (2) (2002) 113–133.
[50] A. D'Elia, D. Piccolo, Maximum likelihood estimation of ARFIMA models with a Box-Cox transformation, Stat. Methods Appl. 12 (3) (2003) 259–275.
[51] P. Omtzigt, P. Paruolo, Impact factors, J. Econometrics 128 (1) (2005) 31–68.
[52] L. Gil-Alana, Measuring the memory parameter on several transformations of asset returns, Int. J. Theor. Appl. Finance 8 (6) (2005) 675–691.
[53] K.a. Ko, M.b. Vannucci, Bayesian wavelet analysis of autoregressive fractionally integrated moving-average processes, J. Statist. Plann. Inference 136 (10) (2006) 3415–3434.
[54] K. Ko, M. Vannucci, Bayesian wavelet-based methods for the detection of multiple changes of the long memory parameter, IEEE Trans. Signal Process. 54 (11) (2006) 4461–4470.
[55] P. Fryzlewicz, G. Nason, Haar-Fisz estimation of evolutionary wavelet spectra, J. R. Stat. Soc. Ser. B Stat. Methodol. 68 (4) (2006) 611–634.
[56] N.a. Haldrup, M.A.b. Nielsen, Estimation of fractional integration in the presence of data noise, Comput. Statist. Data Anal. 51 (6) (2007) 3100–3114.
[57] A.E. Brockwell, Likelihood-based analysis of a class of generalized long-memory time series models, J. Time Series Anal. 28 (3) (2007) 386–407.
[58] W. Palma, Long-Memory time series: theory and methods, 2006.
[59] G. Cavaliere, A. Taylor, Heteroskedastic time series with a unit root, Econ. Theory 25 (5) (2009) 1228–1276.
[60] B. Otok, Suhartono, Development of rainfall forecasting model in Indonesia by using ASTAR, transfer function, and ARIMA methods, Eur. J. Sci. Res. 38 (3) (2009) 386–395.
[61] C. Gourieroux, J. Jasiak, Nonlinear persistence and copersistence, 2010.
[62] J. Pai, N. Ravishanker, Fast bayesian estimation for VARFIMA processes with stable errors, J. Stat. Theory Pract. 4 (4) (2010) 663–677.
[63] D. Percival, D. Mondal, A wavelet variance primer, Handbook of Statist. 30 (2012) 623–657.
[64] J.-P. Chilès, P. Delfiner, Geostatistics: modeling Spatial Uncertainty: Second Edition, 2012.
[65] G. Ljung, J. Ledolter, B. Abraham, George Box's contributions to time series analysis and forecasting, Appl. Stoch. Models Bus. Ind. 30 (1) (2014) 25–35.
[66] A.H. Nury, K. Hasan, M.J.B. Alam, Comparative study of wavelet-ARIMA and wavelet-ANN models for temperature time series data in northeastern Bangladesh, J. King Saud Univ. - Sci.. 29 (1) (2017) 47–61.
[67] R. Douc, K. Fokianos, E. Moulines, Asymptotic properties of quasi-maximum likelihood estimators in observation-driven time series models, Electron. J. Stat. 11 (2) (2017) 2707–2740.
[68] C. Xie, A. Bijral, J. Ferres, NonSTOP: A nonstationary online prediction method for time series, IEEE Signal Process. Lett. (2018).
[69] L.A. Gil-Alana, Modelling the US real GNP with fractionally integrated techniques, Appl. Econ. 36 (8) (2004) 873–879.
[70] G.M.a. Caporale, L.A.b. Gil-Alana, Nonlinearities and fractional integration in the US unemployment rate, Oxf. Bull. Econ. Stat. 69 (4) (2007) 521–544.
[71] F. Corona, G. González-Farías, P. Orraca, A dynamic factor model for the Mexican economy: are common trends useful when predicting economic activity?, Latin Am. Econ. Rev. 26 (1) (2017).
[72] G. Caporale, M. Skare, Long memory in UK real gdp, 1851-2013: An arfima figarch analysis, Transform. Bus. Econ. 17 (1) (2018) 255–268.

[73] P. Fryzlewicz, T. Sapatinas, S. Rao, A eHaar-Fisz technique for locally stationary volatility estimation, Biometrika 93 (3) (2006) 687–704.
[74] N. Gospodinov, A. Gavala, D. Jiang, Forecasting volatility, J. Forecast. 25 (6) (2006) 381–400.
[75] D.F. Hendry, Robustifying forecasts from equilibrium-correction systems, J. Econometrics 135 (1–2) (2006) 399–426.
[76] N. Zhang, A. Lin, P. Shang, Multidimensional k-nearest neighbor model based on EEMD for financial time series forecasting, Physica A 477 (2017) 161–173.
[77] G. Rodriguez, Selecting between autoregressive conditional heteroskedasticity models: An empirical application to the volatility of stock returns in Peru, Rev. Anal. Econ. 32 (1) (2017) 69–94.
[78] M. Ismail, A. Awajan, A new hybrid approach EMD-EXP for short-term forecasting of daily stock market time series data, Electron. J. Appl. Stat. Anal. 10 (2) (2017) 307–327.
[79] G. Ngene, A. Mungai, A. Lynch, Long-term dependency structure and structural breaks: evidence from the U.S. Sector returns and volatility, Rev. Pacific Basin Financ. Markets Policies 21 (2) (2018).
[80] A. Milionis, The importance of variance stationarity in economic time series modelling. A practical approach, Appl. Financ. Econ. 14 (4) (2004) 265–278.
[81] F. Corona, P. Poncela, E. Ruiz, Determining the number of factors after stationary univariate transformations, Empir. Econ. 53 (1) (2017) 351–372.
[82] L.a.A. Gil-Alana, L.b. Jiang, The purchasing power parity hypothesis in the us-china relationship: Fractional integration, time variation and data frequency, Int. J. Finance Econ. 18 (1) (2013) 82–92.
[83] K. Minu, M. Lineesh, C. Jessy John, Wavelet neural networks for nonlinear time series analysis, Appl. Math. Sci. 4 (49–52) (2010) 2485–2495.
[84] O.a. Claveria, S.b. Torra, Forecasting tourism demand to Catalonia: Neural networks vs. time series models, Econ. Modell. 36 (2014) 220–228.
[85] R.M.a.c. Brandao, A.M.O.P.b.d. Nova, Analysis of nonstationary stochastic simulations using classical time-series models, ACM Trans. Model. Comput. Simul. 19 (2) (2009).
[86] C. James, H.A. Murthy, Decoupling non-stationary and stationary components in long range network time series in the context of anomaly detection, in: Proceedings - Conference on Local Computer Networks, LCN, 2012, pp. 76–84.
[87] J.a.b. Gao, Z.a. Leng, B.a. Zhang, X.c. Liu, G.b. Cai, Traffic flow forecasting based on wavelet neural network optimized by GA, in: Chinese Control Conference, CCC, 2013, pp. 8708–8712.
[88] H. Xiong, P. Shang, S. Bian, Detecting intrinsic dynamics of traffic flow with recurrence analysis and empirical mode decomposition, Physica A 474 (2017) 70–84.
[89] L. Barba, N. Rodríguez, A novel multilevel-svd method to improve multistep ahead forecasting in traffic accidents domain, Comput. Intell. Neurosci. 2017 (2017).
[90] H. Zhang, X. Wang, J. Cao, M. Tang, Y. Guo, A multivariate short-term traffic flow forecasting method based on wavelet analysis and seasonal time series, Appl. Intell. 48 (10) (2018) 3827–3838.
[91] U. Sarma, S. Sinha, T. Basu, Medium range forecasting of power system load (energy) demand, Int. J. Syst. Sci. 18 (9) (1987) 1691–1702.
[92] T.K. Bhattacharya, T.K. Basu, Medium range forecasting of monthly energy demand via walsh transform, Int. J. Syst. Sci. 23 (3) (1992) 297–310.
[93] S. Liu, S. Gu, J. Peng, Self-adaptive processing and forecasting algorithm for univariate linear time series, Chin. J. Electron. 26 (6) (2017) 1147–1153.
[94] A. Jamalmanesh, M. Mashhadi, A. Seifi, M. Falahi, Prediction of hydropower energy price using gómes-maravall seasonal model, Int. J. Energy Econ. Policy 8 (2) (2018) 81–88.
[95] N. Bokde, A. Feijóo, K. Kulat, Analysis of differencing and decomposition preprocessing methods for wind speed prediction, Appl. Soft Comput. J. 71 (2018) 926–938.
[96] B. Abraham, N. Balakrishna, Ch. 29. Time series in industry and business, Handbook of Statist. 22 (2003) 1055–1106.
[97] R. Li, Z. Liu, H. Wu, R. Li, G. Dong, J. Jiang, Hierarchical decomposition method and combination forecasting scheme for access load on public map service platforms, Future Gener. Comput. Syst. 87 (2018) 213–227.
[98] A. Lendasse, E. Oja, O. Simula, M. Verleysen, Time series prediction competition: The CATS benchmark, Neurocomputing 70 (13–15) (2007) 2325–2329.
[99] NN3, The nn3 Competition, Technical Report, 2007, http://www.neural-forecasting-competition.com/NN3/index.htm.
[100] NN5, The nn5 Competition, Technical Report, 2008, http://www.neural-forecasting-competition.com/NN5/index.htm.
[101] Ipea, Ipeadata. macroeconomic and regional data, Technical Report, 2017, http://www.ipeadata.gov.br.
[102] R. Salles, L. Assis, G. Guedes, E. Bezerra, F. Porto, E. Ogasawara, A framework for benchmarking machine learning methods using linear models for univariate time series prediction, in: 2017 International Joint Conference on Neural Networks (IJCNN), 2017, pp. 2338–2345.
[103] A. Davydenko, R. Fildes, Measuring forecasting accuracy: the case of judgmental adjustments to sku-level demand forecasts, Int. J. Forecast. 29 (3) (2013) 510–522.
[104] R.P. Salles, E. Ogasawara, TSPred: Functions for Benchmarking Time Series Prediction Prediction, Technical Report, 2018, https://CRAN.R-project.org/package=TSPred.
[105] R.J. Hyndman, Y. Khandakar, Automatic time series forecasting: the forecast package for R, J. Stat. Softw. 27 (3) (2008) 1–22.
[106] D. Kim, H.-S. Oh, EMD: A package for empirical mode decomposition and hilbert spectrum, R J. 1 (1) (2009) 40–46.
[107] E. Aldrich, wavelets: A package of functions for computing wavelet filters, wavelet transforms and multiresolution analyses, Technical Report, 2013, https://cran.r-project.org/web/packages/wavelets/index.html.
[108] R.J. Hyndman, G. Athanasopoulos, Forecasting: Principles and Practice, OTexts, S.l., 2013.
[109] K.E. Taylor, Summarizing multiple aspects of model performance in a single diagram, J. Geophys. Res.: Atmos. 106 (D7) (2001) 7183–7192.